COMPUTER SCIENCE
Complete SLO Solutions

FBISE | SSC-II | Class 10

241 short answers + 99 long answers
All 17 Summative for Theory SLOs

Short questions: 3 marks each
Long questions: 6 marks each
MCQs in the examination: 12

Examination pattern above is used as specified by Fakhar Abbas for this edition. This volume answers the short and long question bank; it does not add a separate MCQ section.

Simple English. Complete working. Clear tables and diagrams.

Prepared by **Fakhar STEM Sphere**

How to use these solutions

Find a question using the same S001-S241 and L001-L099 numbers as the original question bank. Each question is repeated before its model answer. The contents page provides links to every SLO section.

Updated marks

The original bank suggested 2-mark short questions and 5-mark long questions. This solution edition supersedes those suggestions with **3 marks for each short question and 6 marks for each long question**, as requested. Old proposed mark splits have been removed.

Writing answers in the examination

Short answers give the definition, relevant points, example or working needed by the question. Long answers cover every requested part using headings, comparisons, stages or complete calculations. The appropriate length depends on the question; a calculation is assessed through its method and result, rather than a fixed number of sentences.

Practise understanding the answer and then writing it in your own clear words. Reproduce labelled diagrams, calculation steps, code or trace tables when the question requires them. State units and assumptions where relevant.

Scope

Only the 17 SLOs marked Summative for Theory in the supplied assessment framework are covered. The PBA-only SLOs CS-10-C-02 to CS-10-C-05 and EN-10-H-02 remain excluded. Written database queries and language-independent algorithms are included because they fall under theory SLOs.

Accuracy notes

In L057, both models have MSE = 2; the correct conclusion is a tie on the supplied data. Standard illustrative SQL is used; individual database tools may vary in syntax. In authentication answers, recognizing spoken words is distinguished from verifying the speaker's identity.

These are complete practice model answers, not an official marking scheme or a guarantee of full marks. Actual credit depends on the examination question and the examiner's marking scheme.

Sources used

The supplied 13-page Computer Science Grade-X assessment framework and the previously created Theory SLO Question Bank define the scope and questions. The previously supplied solved-paper compilation informed the examination style. No additional formative or practical-only questions have been introduced.

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CS-10-A-01
Number systems and data representation

Theory scope: Number systems; text encoding; binary arithmetic; signed values; instruction sets and addressing modes.

Short questions | 3 marks each

S001. Define a bit and a byte. State the relationship between them. [3 marks]

1. A bit is one binary digit: 0 or 1.

2. A byte is a group of 8 bits.

3. Example: 01000001 contains 8 bits, so it occupies one byte.

S002. Why do computers represent data in binary? Give two reasons. [3 marks]

1. Electronic circuits can easily represent two states, such as OFF and ON.

2. These states represent binary digits 0 and 1 reliably.

3. All instructions and data can therefore be stored and processed as patterns of bits.

S003. State the base and allowed digits of binary, octal, decimal and hexadecimal number systems. [3 marks]

Number system	Base	Allowed digits
Binary	2	0, 1
Octal	8	0 to 7
Decimal	10	0 to 9
Hexadecimal	16	0 to 9 and A to F

A to F represent decimal values 10 to 15.

S004. Convert (45)10 into binary. Show your working. [3 marks]

45 = 32 + 8 + 4 + 1.

Using positions 32, 16, 8, 4, 2, 1 gives digits 1, 0, 1, 1, 0, 1.

Therefore (45)10 = (101101)2.

S005. Convert (101101)2 into decimal. Show the place values used. [3 marks]

(101101)2 = 1x32 + 0x16 + 1x8 + 1x4 + 0x2 + 1x1.

= 32 + 8 + 4 + 1 = 45.

Therefore (101101)2 = (45)10.

S006. Convert (7B)16 into binary and octal. [3 marks]

Hex digits: 7 = 0111 and B = 1011.

Binary: (7B)16 = (01111011)2.

For octal, group from the right: 001 111 011 = 1, 7, 3.

Therefore (7B)16 = (173)8.

S007. Convert (156)8 into decimal and binary. [3 marks]

Decimal: 1x8x8 + 5x8 + 6 = 64 + 40 + 6 = 110.

Binary: replace each octal digit by three bits: 1 = 001, 5 = 101, 6 = 110.

Thus (156)8 = (110)10 = (001101110)2, or (1101110)2 without leading zeros.

S008. Add (1011)2 and (0110)2. Show the carries. [3 marks]

Add from the right: 1+0=1; 1+1=10, so write 0 and carry 1; 0+1+1=10, so write 0 and carry 1; 1+0+1=10, so write 0 and carry 1.

Thus (1011)2 + (0110)2 = (10001)2.

Check: 11 + 6 = 17.

S009. Subtract (0101)2 from (1100)2 using binary subtraction. [3 marks]

From the right, 0-1 needs borrowing. The borrow passes through the next 0 from the 4's position.

The four result bits are 0, 1, 1, 1.

Thus (1100)2 - (0101)2 = (0111)2. Check: 12 - 5 = 7.

S010. Find the 1's complement and 2's complement of the 8-bit number 00101101. [3 marks]

Original: 00101101.

1's complement: invert every bit, giving 11010010.

2's complement: add 1 to the 1's complement: 11010010 + 1 = 11010011.

S011. Differentiate between signed and unsigned binary numbers. [3 marks]

Unsigned	Signed 2's complement
Represents zero and positive integers.	Represents negative, zero and positive integers.
All bits contribute to a non-negative value.	The leftmost bit has a negative place value.
8-bit range: 0 to 255.	8-bit range: -128 to 127.

S012. Write the range of an 8-bit unsigned integer and an 8-bit signed 2's-complement integer. [3 marks]

Unsigned range: 0 to 2^8 - 1 = 0 to 255.

Signed 2's-complement range: -2^7 to 2^7 - 1 = -128 to 127.

The same 8 bits have different interpretations depending on the representation used.

S013. What is overflow? Explain using the 4-bit unsigned addition 1111 + 0001. [3 marks]

Overflow occurs when a result is outside the range of the chosen number representation.

For 4-bit unsigned numbers, the maximum is 15.

1111 + 0001 = 10000, or decimal 16. This needs 5 bits, so a 4-bit result overflows and retains only 0000.

S014. What is underflow in the storage of very small non-zero real numbers? [3 marks]

Underflow occurs when a non-zero real result is too small in magnitude for the normal range of the storage format.

It may be stored with reduced precision, or rounded to zero.

Example: repeated division of a very small floating-point number may cause underflow.

S015. Define an encoding scheme. Why are standard character codes needed? [3 marks]

An encoding scheme assigns codes to characters or other data.

Standard codes allow different computers to store and exchange text consistently.

Examples are ASCII and Unicode; without an agreed code, a received number may be interpreted as the wrong character.

S016. Give two differences between ASCII and Unicode. [3 marks]

ASCII	Unicode
Original ASCII uses 7-bit codes for 128 characters.	Provides codes for characters from many languages.
Mainly supports basic English letters, digits and symbols.	Supports English, Urdu and many other scripts.

Unicode is encoded using formats such as UTF-8; it does not use one fixed bit length for all text.

S017. The ASCII code of A is decimal 65. Write its 8-bit binary code. [3 marks]

65 = 64 + 1.

In 8-bit positions 128, 64, 32, 16, 8, 4, 2, 1, this is 01000001.

Therefore ASCII A = (65)10 = (01000001)2.

S018. Define an instruction set and an addressing mode. [3 marks]

An instruction set is the collection of operations a processor can execute, such as ADD, LOAD and STORE.

An addressing mode tells how an instruction obtains its operand.

Examples: an immediate operand is given in the instruction; a direct-memory operand is found at a stated memory address.

CS-10-A-01 | Long answers

Long questions | 6 marks each

L001. Convert (3A7)16 into binary and octal. Show each conversion step. [6 marks]

Step 1 - Hexadecimal to binary:

Hexadecimal digit	Four-bit binary
3	0011
A	1010
7	0111

Therefore (3A7)16 = (001110100111)2.

Step 2 - Binary to octal: group the same bits into threes from the right: 001 110 100 111.

These groups are 1, 6, 4 and 7. Therefore (3A7)16 = (1647)8.

Check: hexadecimal value = 3x256 + 10x16 + 7 = 935. Octal value = 1x512 + 6x64 + 4x8 + 7 = 935.

L002. Represent +25 and -25 in 8-bit 2's-complement form. Use 2's-complement addition to calculate 42 - 25 and interpret the result. [6 marks]

1. Write +25 in 8 bits: 00011001.

2. Invert the bits: 11100110. Add 1 to obtain -25: 11100111.

3. Write +42 in 8 bits: 00101010.

4. Add +42 and -25:

```
  00101010
+ 11100111
-----------
1 00010001
```

5. Discard the carry beyond 8 bits. The result is 00010001.

6. Its leftmost bit is 0, so it is positive: 16 + 1 = 17. Therefore 42 - 25 = 17, with no signed overflow.

L003. Explain binary addition and overflow. Calculate 0111 + 0001 as 4-bit unsigned numbers and as 4-bit signed 2's-complement numbers; explain why the interpretations differ. [6 marks]

Binary addition uses 0+0=0, 0+1=1, and 1+1=10, where 1 is carried to the next position.

```
  0111
+ 0001
------
  1000
```

Unsigned interpretation: 0111 = 7 and 0001 = 1. The result 1000 = 8 lies within 0 to 15. There is no unsigned overflow.

Signed 2's-complement interpretation: both inputs are positive, but 1000 represents -8. The correct result +8 is outside the 4-bit signed range -8 to +7, so signed overflow occurs.

Conclusion: overflow depends on the representation, not just the bit pattern.

L004. Explain how text is represented using character codes. Compare ASCII and Unicode, and explain why Unicode is useful for English and Urdu text. [6 marks]

1. Each character is assigned a numerical code by a character standard.

2. The code is stored and transmitted in binary using an encoding format.

3. When text is displayed, the computer interprets the stored codes and displays the corresponding characters.

4. Original ASCII has 128 codes and uses 7 bits. It represents basic English letters, digits, punctuation and control characters.

5. Unicode provides characters for many languages and scripts. Formats such as UTF-8 encode these characters as bytes.

6. ASCII cannot represent Urdu letters. Unicode can represent both English and Urdu, so it is suitable for a bilingual document.

L005. Explain data representation, an instruction set and addressing modes. Give a simple example of an instruction that uses an immediate value and one that uses a memory address. [6 marks]

Data representation: computers encode numbers, text and instructions as binary patterns. For example, decimal 5 can be stored as 00000101.

Instruction set: the operations supported by a processor, such as LOAD, ADD and STORE. Programs must ultimately use instructions compatible with the processor.

Addressing modes: rules for finding an instruction's operand.

Immediate example: LOAD R1, #5 means put the value 5 directly into register R1.

Direct-memory example: LOAD R1, [1000] means load the value stored at memory address 1000 into R1.

These are illustrative assembly-style instructions; exact notation depends on the processor.

L006. Explain bits, bytes and signed and unsigned numbers. State the unsigned and signed 2's-complement ranges for 8 bits and explain the difference. [6 marks]

1. A bit is a binary digit, 0 or 1. A byte contains 8 bits.

2. Unsigned integers use all bits to represent non-negative values.

3. With 8 bits, unsigned values range from 0 to 2^8 - 1 = 255.

4. Signed 2's-complement integers represent negative and positive values. The leftmost bit has weight -128; the remaining bits have weights 64 to 1.

5. Their 8-bit range is -2^7 to 2^7 - 1, or -128 to 127.

6. For example, 11111111 represents 255 unsigned, but -1 in signed 2's complement. The same bits therefore require a stated interpretation.

CS-10-A-02
Operating systems

Theory scope: OS services; process and thread management; scheduling; synchronization; interrupts; file systems; memory and virtual memory.

Short questions | 3 marks each

S019. Define an operating system. Give two examples. [3 marks]

An operating system (OS) is system software that manages computer hardware and provides services to applications.

It coordinates resources such as CPU time, memory and files.

Examples: Windows and Linux.

S020. State two resources managed by an operating system. [3 marks]

1. CPU: the OS schedules processes and shares processing time.

2. Main memory: it allocates RAM to programs and releases it when no longer needed.

It also manages files, storage devices and input/output devices.

S021. How does an operating system help an application communicate with hardware? [3 marks]

The application requests an OS service, often through a system call.

The OS uses a device driver to communicate with the relevant hardware.

For example, a word processor asks the OS to print, and the printer driver sends suitable commands to the printer.

S022. What is a system call? Give one example of a service requested through it. [3 marks]

A system call is a controlled request from a program to the operating system for a service.

Example: requesting that a file be opened or read.

The OS checks the request and performs the protected operation on the program's behalf.

S023. Differentiate between a process and a thread. [3 marks]

Process	Thread
A program that is currently executing.	An execution path within a process.
Normally has its own memory space.	Shares its process's memory and resources.
Example: a running web browser process.	Example: a thread handling a task in that process.

S024. What is process scheduling? Why is it required? [3 marks]

Process scheduling is the OS activity of choosing which ready process gets CPU time next.

It allows several programs to share the CPU.

It improves responsiveness and prevents one ordinary process from occupying the processor indefinitely.

S025. What is synchronization? Give one situation where it is needed. [3 marks]

Synchronization coordinates processes or threads that use shared resources.

It prevents conflicting actions and inconsistent data.

Example: if two threads update the same bank balance, access must be coordinated so one update does not overwrite the other.

S026. Define deadlock and give a simple example involving two resources. [3 marks]

Deadlock occurs when processes wait for resources held by each other and cannot continue.

Example: A holds a printer and waits for a scanner; B holds the scanner and waits for the printer.

Neither can proceed unless the situation is broken.

S027. What is an interrupt? Why is it useful? [3 marks]

An interrupt is a signal that asks the processor to handle an event.

The processor temporarily changes its execution to an interrupt handler, then can resume its previous work.

Example: a keyboard or storage device signals that input or an operation is ready.

S028. State two functions of file management. [3 marks]

1. The OS creates, opens, saves, renames and deletes files.
2. It organizes files into folders and tracks their storage locations.
3. It also controls access permissions so users can access only authorized files.

S029. What is a file system? Explain the purpose of folders. [3 marks]

A file system is the method an OS uses to name, organize and store files on a storage device.

Folders group related files and may contain subfolders.

For example, a School folder can contain separate Computer Science and Mathematics folders.

S030. State two functions of memory management. [3 marks]

1. The OS tracks which RAM locations are free and in use.
2. It allocates memory to programs and releases memory after use.
3. It protects one program's memory from unauthorized access by another program.

S031. Differentiate between main memory and virtual memory. [3 marks]

Main memory is physical RAM used for currently active programs and data.

Virtual memory is a memory-management system that gives processes a virtual address space; it can use storage to support data not currently in RAM.

Accessing stored pages is much slower than accessing RAM.

S032. Why may a computer slow down when many applications are open? [3 marks]

Many applications compete for CPU time and RAM.

When RAM is insufficient, frequent transfers between RAM and storage can slow the computer considerably.

Closing unused applications reduces the load; adding RAM may help when memory is the limiting resource.

S033. Differentiate between single-user and multi-user operating systems. [3 marks]

A single-user OS is designed mainly for one user at a time, although that user may run many applications.

A multi-user OS supports multiple users with separate accounts and controlled access, often simultaneously.

The OS must share resources and protect each user's work.

S034. What is a real-time operating system? Give two applications. [3 marks]

A real-time OS manages tasks so they meet required timing deadlines.

Examples include an aircraft flight-control system and an industrial robot controller.

Correct timing matters because a response delivered too late may be unsafe or useless.

S035. How do device drivers help with input and output? [3 marks]

A device driver is software that allows the OS to control a particular device.

It converts general OS requests into device-specific commands.

For example, a keyboard driver supports input, while a printer driver supports printed output.

S036. State the main stages in a process's life, from creation to completion. [3 marks]

New: the process is created. Ready: it waits for CPU time. Running: its instructions execute.

Waiting: it waits for input/output or another event, then returns to ready.

Terminated: it finishes and its resources are released.

CS-10-A-02 | Long answers

Long questions | 6 marks each

L007. Define an operating system. Explain its file management and memory management functions with examples. [6 marks]

Significance: an OS provides an environment for applications and coordinates hardware resources. Without it, each application would need to manage many low-level operations itself.

File management: it creates, opens, saves, copies, renames and deletes files. It organizes folders, tracks storage locations and controls permissions. Example: saving a report inside a subject folder.

Memory management: it tracks RAM use, allocates memory to programs, releases unused memory and protects each process's memory. It may use virtual memory when required.

Example: when a browser and a word processor run together, the OS gives each memory and prevents one from directly changing the other's protected data.

L008. Explain how an operating system manages input, output, storage, processing and running applications. [6 marks]

1. Input management: the OS receives data from devices such as the keyboard through drivers and passes it to the correct application.

2. Output management: it sends results to monitors, printers and other devices in a suitable form.

3. Storage management: it organizes files, controls reading and writing, and keeps track of storage space.

4. Processing management: it schedules processes and shares CPU time among them.

5. Memory management: it allocates RAM and protects programs' memory areas.

6. Application support: it loads and starts programs, handles service requests and releases resources when they finish.

Example: opening, editing and printing a document uses all these services.

L009. Explain process scheduling and synchronization. Describe a deadlock involving a printer and scanner. [6 marks]

Scheduling: the OS selects a ready process to run and gives it CPU time. Sharing time allows several applications to remain responsive.

Synchronization: processes or threads coordinate access to shared data or devices. For example, a lock can allow only one thread to update shared data at a time.

Deadlock example:

1. Process A holds the printer and requests the scanner.

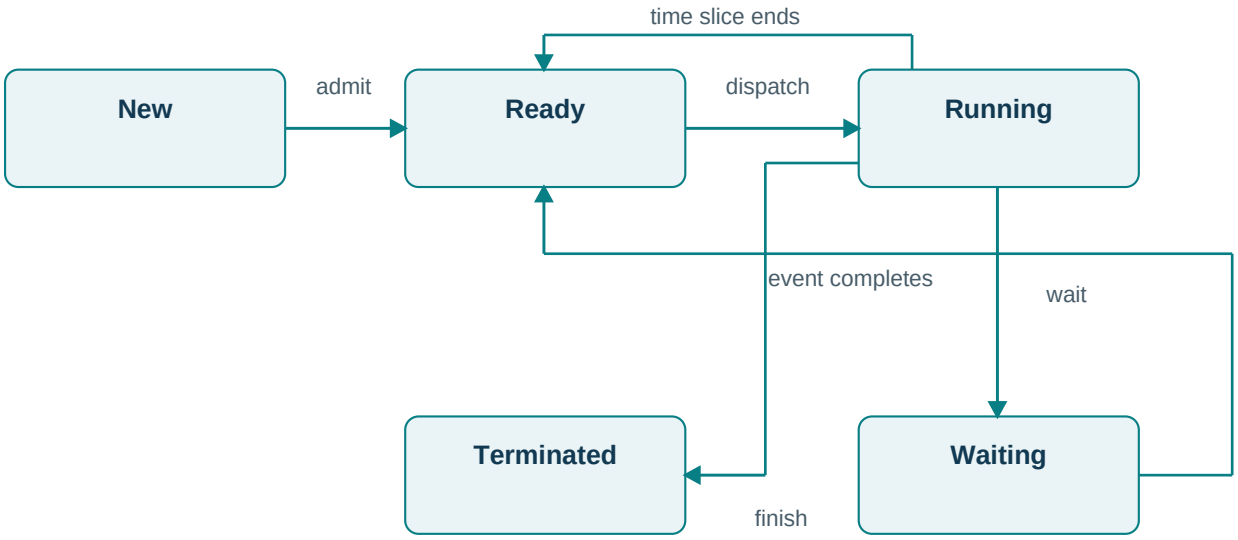
2. Process B holds the scanner and requests the printer.

3. Each waits for the resource held by the other, so neither progresses.

A consistent order for acquiring resources can help prevent this situation. Scheduling chooses who runs; synchronization controls safe sharing.

L010. Describe process states: new, ready, running, waiting and terminated. Show possible transitions in a labelled diagram. [6 marks]

State	Meaning
New	The OS is creating the process.
Ready	The process can run but is waiting for CPU time.
Running	The CPU executes the process's instructions.
Waiting	The process is waiting for an event, such as input/output.
Terminated	Execution has finished and resources can be released.



A running process may return to ready when its time slice ends. When an awaited event finishes, a waiting process returns to ready rather than directly taking the CPU.

L011. Compare batch, time-sharing and real-time operating systems. Give a suitable use of each. [6 marks]

Type	Main idea	Suitable use
Batch OS	Executes submitted jobs in batches with little interaction during execution.	Processing payroll records.
Time-sharing OS	Shares CPU time among interactive processes or users.	A shared university computer server.
Real-time OS	Schedules work to meet required response deadlines.	Industrial machine control.

Batch processing emphasizes completing groups of jobs. Time sharing emphasizes interactive response. Real-time operation emphasizes meeting timing requirements; a late response may count as a failure even if the calculation is correct.

L012. A student runs a browser, word processor and video editor together. Explain CPU scheduling, RAM allocation and virtual memory in this situation. Suggest two ways to improve performance. [6 marks]

CPU scheduling: the OS divides CPU time among the browser, word processor and video editor so that each can progress.

RAM allocation: it gives each application memory for its instructions and data and protects those areas.

Virtual memory: inactive pages may be kept on storage and loaded when needed. Excessive paging can make the computer slow because storage access is slower than RAM.

Improvement 1: close unnecessary applications and background tasks to reduce CPU and memory demand.

Improvement 2: add RAM if memory is regularly insufficient. A powerful processor helps only if CPU workload is also a limiting factor.

L013. Explain how a program written in a programming language runs on computer hardware, referring to translation, the operating system and the processor's instruction set. [6 marks]

1. A programmer writes source code in a language such as C++.

2. A translator converts it into a form suitable for execution. For compiled programs, the output ultimately contains machine instructions.

3. The generated instructions must match the processor's instruction set and the target system.

4. The OS loads the program into memory and creates a process.

5. It schedules CPU time, allocates memory and provides input/output and file services.

6. The processor fetches, decodes and executes the machine instructions. When the program finishes, the OS releases its resources.

Thus the language expresses the solution, translation prepares executable instructions, and the OS coordinates execution.

CS-10-A-03
Software and development tools

Theory scope: Software categories; productivity software; translators; editors; IDEs; debuggers; repositories; build systems; online, offline and hosted software.

Short questions | 3 marks each

S037. Differentiate between system software and application software, with examples. [3 marks]

System software manages hardware and provides a platform for applications. Examples: Windows and device drivers.

Application software performs user tasks. Examples: a word processor and web browser.

Applications rely on system software to access memory, files and devices.

S038. What is driver software? Give one example. [3 marks]

Driver software allows an OS to communicate with particular hardware.

Example: a printer driver converts printing requests into commands understood by the printer.

The correct driver helps the device operate properly.

S039. What is programming software? Name two tools used by programmers. [3 marks]

Programming software helps developers create, translate and check programs.

Examples: a source-code editor for writing code and a compiler for translating it.

A debugger is another tool that helps locate errors during execution.

S040. Differentiate between a compiler and an interpreter. [3 marks]

Compiler	Interpreter
Translates source code before executing the resulting program.	Executes a program through an interpreting runtime, often statement by statement.
Commonly produces object or executable code.	Does not necessarily create a separate native executable.
Example: a C++ compiler.	Example: a Python interpreter.

Some modern language systems combine compilation and interpretation.

S041. What is an assembler? What kind of language does it translate? [3 marks]

An assembler translates assembly language into machine code.

Assembly uses symbolic instructions such as ADD and LOAD.

The generated machine instructions must match the intended processor architecture.

S042. Define an IDE. Name two tools it usually contains. [3 marks]

An IDE, or Integrated Development Environment, combines programming tools in one application.

Two common components are a source-code editor and a debugger.

It may also provide a compiler, build tools and a way to run programs.

S043. Differentiate between a source-code editor and a debugger. [3 marks]

A source-code editor is used to write and modify program text, often with syntax highlighting.

A debugger helps examine a running program and locate errors.

For example, an editor changes a formula, while a debugger helps discover why its result is wrong.

S044. What is a code repository? State one benefit of using it. [3 marks]

A code repository stores a project's source code and related files.

With version control, it records changes over time.

This helps developers recover earlier work and coordinate contributions.

S045. How does version control help a team of programmers? Give two points. [3 marks]

Version control records who changed files and when, and allows comparison with earlier versions.

Branches allow team members to work on separate features.

Reviewed changes can be merged, while conflicts can be identified and resolved.

S046. What is a build system? State its purpose. [3 marks]

A build system automates the steps needed to turn source files into a usable program.

It may run compilation, linking and related tasks in the correct order.

It can rebuild only changed parts and their dependencies, saving development time.

S047. Differentiate between on-premises and cloud-hosted applications. [3 marks]

On-premises software is hosted on an organization's own computers or servers.

Cloud-hosted software runs on infrastructure provided remotely and is usually accessed over a network.

They differ in responsibility for maintenance, access arrangements and control of the hosting environment.

S048. Give two benefits and two limitations of online computing platforms. [3 marks]

Benefits: users can access work remotely, and teams can collaborate on shared projects.

Limitations: reliable internet may be needed, and service outages can interrupt work.

Users must also consider access permissions and privacy when storing information online.

S049. State two situations where an offline application is useful. [3 marks]

1. Offline software is useful where internet access is unavailable or unreliable.

2. It allows local work during a network outage or while travelling.

Example: a locally installed word processor can edit a downloaded document without an internet connection.

S050. Name suitable productivity software for writing a report and analysing numerical data. [3 marks]

A word processor, such as Microsoft Word, is suitable for writing and formatting a report.

A spreadsheet, such as Microsoft Excel, is suitable for calculations and numerical analysis.

The spreadsheet can use formulas, tables and charts to summarize data.

S051. A new printer is connected but does not work properly. Which type of software may be needed, and why? [3 marks]

A suitable printer driver may be needed.

The driver allows the operating system to send commands and data in a form the printer understands.

Installing the correct driver for the printer and OS can solve a driver-related communication problem.

CS-10-A-03 | Long answers

Long questions | 6 marks each

L014. Describe five software-development tools: a translator, editor, debugger, IDE and code repository. State the purpose of each. [6 marks]

Tool	Purpose	Example
Translator	Converts source code into another executable or lower-level form.	A C++ compiler.
Source-code editor	Writes and edits source files.	Notepad++.
Debugger	Examines execution to help locate errors.	GDB.
IDE	Combines editing, running, building and debugging tools.	Visual Studio.
Code repository	Stores and organizes project code.	A Git repository hosted on GitHub.

Together, these tools support the development cycle: write code, translate or run it, examine problems and maintain the project's history. A repository used with version control also supports teamwork and recovery.

L015. Compare system, application, programming and driver software. Give one example of each and explain how they work together on a computer. [6 marks]

Category	Purpose	Example
System software	Manages resources and supports other software.	An operating system such as Windows.
Application software	Performs a user's task.	A word processor.
Programming software	Helps create and check programs.	A compiler or debugger.
Driver software	Connects the OS to specific hardware.	A printer driver.

Example of cooperation: a developer writes a program using programming software. The resulting application runs under the OS. If it prints a report, the OS uses a printer driver to operate the printer.

Driver software is commonly treated as a type of system software.

L016. Compare compilers, interpreters and assemblers, referring to their input, purpose and way of translation. [6 marks]

Tool	Input	Main purpose	Translation approach
Compiler	High-level source code.	Produces lower-level code, often object or machine code.	Translates before the resulting code runs.
Interpreter	Code in an interpreted language or an intermediate form.	Executes the program through a runtime.	Carries out instructions as the program runs.
Assembler	Assembly-language instructions.	Produces processor-specific machine code.	Converts symbolic assembly instructions into binary instructions.

Examples are a C++ compiler, a Python interpreter and an assembler for a particular processor. Actual language implementations may combine techniques, but the basic roles remain distinct.

L017. Explain how code repositories and version control support change tracking, recovery and teamwork. [6 marks]

1. Central organization: a repository keeps source code and related project files together.

2. Change tracking: version control records commits showing what changed and who made the change.

3. Comparison: developers can compare versions to understand modifications.

4. Recovery: earlier versions can be restored if a change causes a problem.

5. Separate work: branches allow developers to work on different features without immediately changing the shared version.

6. Team review: contributions can be reviewed and merged; conflicts are resolved before work is combined.

Example: a school website team can maintain its code in a Git repository hosted on GitHub.

L018. A school must choose between locally installed and cloud-hosted productivity applications. Compare internet needs, access, collaboration, maintenance and control of data. [6 marks]

Factor	Locally installed/on-premises	Cloud-hosted
Internet	Many tasks work without internet.	Access usually depends on internet or a network.
Access	Mainly through installed devices or a managed local network.	Often accessible from multiple locations and devices.
Collaboration	May require a separate sharing system.	Shared editing is often built in.
Maintenance	The school manages local installation and updates.	The provider manages much of the hosted service.
Data control	The school controls its own storage environment.	Data is hosted remotely; permissions and provider arrangements matter.

A school with unreliable internet may favour offline work; one needing frequent remote collaboration may benefit from cloud hosting. The choice should follow actual needs.

L019. Explain the roles of a language editor, compiler, debugger and build system when developing a program. Describe how an IDE combines such tools. [6 marks]

1. A language editor lets the programmer write and save source code. Syntax highlighting helps make it readable.

2. A compiler checks the source and translates it into lower-level code. Compilation errors must be corrected.

3. A debugger helps inspect the program's execution and locate causes of incorrect behaviour.

4. A build system automates compilation, linking and other required steps, including rebuilding changed parts.

5. An IDE combines these tools in one environment so the developer can edit, build, run and debug conveniently.

6. Example workflow: write a program, build it, run it, investigate an incorrect result, edit the code and rebuild.

CS-10-B-01
Counting, logic, searching and sorting

Theory scope: Counting principles; algorithm properties; Boolean, verbal and non-verbal reasoning; linear and binary search; bubble and insertion sort.

Short questions | 3 marks each

S052. State the addition principle of counting and give an example. [3 marks]

The addition principle applies to mutually exclusive choices: if one task can be done in m ways or another in n ways, there are m+n choices.

Example: choosing one drink from 3 juices or 2 milkshakes gives 3+2=5 choices.

The alternatives must not overlap unless overlap is subtracted.

S053. A student has 3 shirts and 4 trousers. How many different shirt-trouser pairs are possible? Name the counting principle. [3 marks]

There are 3 choices of shirt and 4 choices of trousers for each shirt.

By the multiplication principle, total pairs = 3 x 4 = 12.

Each different shirt-trouser combination is counted once.

S054. Differentiate between a permutation and a combination. [3 marks]

A permutation is an arrangement in which order matters; AB and BA are different.

A combination is a selection in which order does not matter; selecting A and B is the same as selecting B and A.

Example: assigning captain and vice-captain uses permutations; choosing a two-person team uses combinations.

S055. In how many ways can 4 different books be arranged on a shelf? [3 marks]

There are 4 choices for the first position, then 3, then 2, then 1.

Total arrangements = 4! = 4 x 3 x 2 x 1 = 24.

Order matters because changing the sequence produces a different arrangement.

S056. In how many ways can a team of 2 be chosen from 5 students? [3 marks]

Choose 2 from 5 without regard to order:

C(5,2) = 5! / (2! x 3!) = (5 x 4)/(2 x 1) = 10.

Dividing by 2 avoids counting each pair in both orders.

S057. State the pigeonhole principle. Apply it to 13 students and 12 birth months. [3 marks]

The pigeonhole principle states that placing more objects than boxes guarantees that at least one box contains two or more objects.

Here, the 13 students are objects and the 12 birth months are boxes.

Therefore at least two students share a birth month.

S058. In a class, 18 students like Python, 12 like JavaScript and 5 like both. How many like at least one of these languages? [3 marks]

Number liking at least one language = number liking Python + number liking JavaScript - number liking both.

= 18 + 12 - 5 = 25.

The overlap is subtracted because those 5 students were counted twice.

S059. Define an algorithm. State any two properties of a good algorithm. [3 marks]

An algorithm is a finite sequence of clear steps for solving a problem.

Definiteness means each instruction is unambiguous.

Finiteness means the procedure finishes after a limited number of steps.

S060. Differentiate between definiteness and finiteness. [3 marks]

Definiteness means that each step has a clear, precise meaning.

Finiteness means that the algorithm terminates after a limited number of steps.

An endless loop may use clear instructions but still fail the finiteness requirement.

S061. What is meant by the effectiveness and generality of an algorithm? [3 marks]

Effectiveness means each step is basic, practical and executable in a finite time.

Generality means the algorithm solves all valid instances of the intended problem, rather than only one fixed example.

For instance, a general averaging algorithm can accept different valid lists of marks.

S062. Find the value of (P AND NOT Q) OR Q when P is true and Q is false. [3 marks]

Given P = true and Q = false, NOT Q = true.

P AND NOT Q = true AND true = true.

Therefore (P AND NOT Q) OR Q = true OR false = true.

S063. Differentiate between AND, OR and NOT using simple statements. [3 marks]

AND is true only when both statements are true: "I have a pen AND paper."

OR is true when at least one statement is true: "I have a pen OR pencil."

NOT reverses the truth value: NOT "It is raining" means "It is not raining."

S064. All tablets are computing devices. Some computing devices are portable. Can we conclude that all tablets are portable? Give a reason. [3 marks]

No. The statement says that only some computing devices are portable.

It does not say whether tablets belong to that portable group.

Therefore the conclusion that all tablets are portable does not follow from the given statements.

S065. Complete the pattern 2, 4, 8, 16, __, __ and state the rule. [3 marks]

The missing values are 32 and 64.

Rule: multiply the previous value by 2.

Thus 16 x 2 = 32 and 32 x 2 = 64.

S066. A shape pattern repeats: circle, square, triangle. Which shape is in position 8? Explain. [3 marks]

The pattern repeats every 3 positions: circle, square, triangle.

8 divided by 3 leaves remainder 2, so position 8 matches the second shape.

Therefore the shape is a square.

S067. Differentiate between searching and sorting. [3 marks]

Searching finds a required item or reports that it is absent.

Sorting rearranges items into an order, such as ascending marks.

Example: linear search finds roll number 25; bubble sort arranges all roll numbers.

S068. Describe linear search. Can it work on unsorted data? [3 marks]

Linear search checks items one by one until the target is found or the list ends.

It works on unsorted data because it does not rely on order.

For a list of n items, an unsuccessful search may check all n items.

S069. State two conditions or requirements for efficient binary search on an array. [3 marks]

1. The array must be sorted according to the comparison used.
2. Items must be comparable so the algorithm can choose the lower or upper half.
3. Efficient access to the middle item is needed; arrays provide this through indexing.

S070. Why is binary search usually faster than linear search on a large sorted array? [3 marks]

Binary search checks a middle item and eliminates half the remaining range after each comparison.

Linear search checks items one by one and may inspect the entire list.

Thus binary search needs far fewer comparisons on a large sorted array: about log2(n), rather than up to n.

S071. What happens during one pass of bubble sort? [3 marks]

Bubble sort compares adjacent items and swaps them if they are in the wrong order.

In ascending sorting, a full pass moves the largest item in the unsorted part to its rightmost position.

Further passes sort the remaining part.

S072. Briefly describe how insertion sort builds a sorted list. [3 marks]

Insertion sort starts with a sorted part containing the first item.

It takes the next item, shifts larger items right and inserts the item in its correct position.

Repeating this process grows the sorted part until the whole list is sorted.

CS-10-B-01 | Long answers

Long questions | 6 marks each

L020. Explain six properties of a good algorithm: input, output, definiteness, finiteness, effectiveness and generality. Include a simple example. [6 marks]

1. Input: the algorithm clearly states its required data; some algorithms need no external input.

2. Output: it produces at least one result.

3. Definiteness: each instruction is precise and unambiguous.

4. Finiteness: it stops after a limited number of steps.

5. Effectiveness: its steps are simple and possible to carry out in finite time.

6. Generality: it works for all valid inputs in its intended problem class.

Example: to add two numbers, input a and b, calculate sum = a+b, output sum and stop. The steps are clear, practical, finite and apply to different valid pairs.

L021. Compare linear and binary search. Find 35 in [5, 10, 15, 20, 25, 30, 35] using both methods and show the comparisons. [6 marks]

Linear search checks one item at a time and can work on unsorted data. Binary search repeatedly halves a sorted search range and needs efficient access to the middle.

Target: 35 in [5, 10, 15, 20, 25, 30, 35].

Linear search compares 5, 10, 15, 20, 25, 30 and 35: 7 comparisons.

Binary step	Search positions, using 1-based positions	Middle value	Decision
1	1 to 7	20 at position 4	Search positions 5 to 7.
2	5 to 7	30 at position 6	Search position 7.
3	7 to 7	35 at position 7	Found.

Binary search uses 3 comparisons here.

L022. Use bubble sort to arrange [7, 3, 5, 2, 4] in ascending order. Show the list after each full pass and explain the stopping condition. [6 marks]

Bubble sort compares neighbours and swaps them when the left value is larger.

Stage	List after the full pass
Initial	7, 3, 5, 2, 4
Pass 1	3, 5, 2, 4, 7
Pass 2	3, 2, 4, 5, 7
Pass 3	2, 3, 4, 5, 7
Pass 4	2, 3, 4, 5, 7 - no swaps

The largest remaining unsorted value moves right during each pass. An optimized version stops when a full pass makes no swaps. A basic version completes n-1 passes. Sorted result: [2, 3, 4, 5, 7].

L023. Use insertion sort to arrange [8, 4, 6, 2, 5] in ascending order. Show the list after each insertion and explain the method. [6 marks]

Insertion sort keeps a sorted left part and inserts the next value in the correct place by shifting larger values right.

Step	Inserted value	List after insertion
Initial	First item 8	8, 4, 6, 2, 5
1	4	4, 8, 6, 2, 5
2	6	4, 6, 8, 2, 5
3	2	2, 4, 6, 8, 5
4	5	2, 4, 5, 6, 8

After each step, the sorted part grows by one item. After the last insertion, all five items are sorted.

L024. A club has 6 students. Calculate the ways to choose a captain and vice-captain, and the ways to choose a 2-student team. Explain why the answers differ. [6 marks]

Captain and vice-captain: these are different roles, so order matters. There are 6 choices of captain and then 5 choices of vice-captain.

P(6,2) = 6 x 5 = 30.

Two-student team: roles are not assigned, so order does not matter.

C(6,2) = (6 x 5)/(2 x 1) = 15.

Reason for the difference: choosing Ali as captain and Sara as vice-captain is different from reversing their roles. But a team containing Ali and Sara is the same team in either order. Therefore each unordered pair corresponds to two ordered role assignments.

L025. In a group, 20 students study art, 15 study music and 8 study both. Find the number studying at least one subject. Then explain why among 13 students at least two share a birth month. [6 marks]

Part A - Inclusion-exclusion:

Number studying art or music = 20 + 15 - 8 = 27.

The 8 students studying both were counted twice, so subtract them once.

Part B - Pigeonhole principle:

There are 12 possible birth months. If each month contained at most one of the 13 students, only 12 students could be placed.

The thirteenth student must share a month with at least one other student.

Therefore at least two of the 13 students have the same birth month. This guarantees a repeated month, not a particular month or date.

L026. Construct truth tables for P AND Q, P OR Q and NOT P. Use them to evaluate (P OR Q) AND NOT Q for all input pairs. [6 marks]

AND is true only when both inputs are true; OR is true when at least one is true; NOT reverses an input.

P	Q	P AND Q	P OR Q	NOT P	NOT Q	(P OR Q) AND NOT Q
F	F	F	F	T	T	F
F	T	F	T	T	F	F
T	F	F	T	F	T	T
T	T	T	T	F	F	F

The final expression is true only when P is true and Q is false. Evaluate the expression in stages: first OR, then NOT Q, then AND the two results.

L027. Write a language-independent algorithm to find a student's roll number in an unsorted list. Explain why your chosen search method is suitable. [6 marks]

Algorithm - linear search, using positions 1 to n:

```
INPUT n, rollNumbers[1..n], target
position = -1
FOR i = 1 TO n
    IF rollNumbers[i] = target THEN
        position = i
        BREAK
    END IF
END FOR
OUTPUT position
```

If position is -1, the roll number is absent. Otherwise, position is its first matching location.

Linear search is suitable because the list is unsorted. It checks items directly without needing to sort the list first. Binary search cannot safely discard halves of an unsorted list.

L028. A password consists of one uppercase English letter followed by three decimal digits. Repetition is allowed. Calculate the number of passwords. Then calculate the total if the three digits cannot repeat. Explain the principles used. [6 marks]

Part A - Repetition allowed:

There are 26 choices for the letter. Each of the three digit positions has 10 choices, including 0.

Total = 26 x 10 x 10 x 10 = 26,000.

Part B - Digits cannot repeat:

There are 26 choices for the letter, 10 for the first digit, 9 for the second and 8 for the third.

Total = 26 x 10 x 9 x 8 = 18,720.

Both use the multiplication principle. The second also uses ordered selection without replacement for the digits. A leading zero is allowed because this is a password, not a three-digit integer.

CS-10-B-02
Abstraction, modular solutions and trace tables

Theory scope: Algorithm design; generalization; modularity; dry runs; syntax and logic errors. Use language-independent pseudocode.

Short questions | 3 marks each

S073. Define abstraction. Give one everyday example. [3 marks]

Abstraction means focusing on essential features while hiding unnecessary details.

Example: a driver uses a car's steering wheel without needing to know every engine operation.

This reduces complexity and makes the system easier to use or understand.

S074. How does abstraction help create a generalized solution? [3 marks]

Abstraction identifies the features shared by similar problems and ignores unnecessary differences.

A solution can then accept inputs instead of using fixed values.

For example, one averaging module can calculate the mean of marks for any class.

S075. Give two advantages of dividing a solution into modules. [3 marks]

1. Each module has a clear task, making the solution easier to understand.

2. Modules can be tested, corrected and reused independently.

3. A change to one module often requires fewer changes elsewhere, making maintenance easier.

S076. Differentiate between an algorithm and a program. [3 marks]

An algorithm is a clear step-by-step method for solving a problem, independent of a programming language.

A program implements a method using a particular language and its syntax.

The same algorithm can therefore be implemented as different programs.

S077. What is a dry run? Why is it useful? [3 marks]

A dry run is manually following an algorithm using sample inputs.

The changing values of variables and the output are recorded.

It helps check the algorithm's logic and locate mistakes before or during implementation.

S078. What information is recorded in a trace table? [3 marks]

A trace table records the values of important variables at each step or iteration.

It can also include condition results, inputs and outputs.

For example, a summing loop may record the loop counter and running total after each iteration.

S079. Differentiate between a syntax error and a logic error. [3 marks]

A syntax error breaks the language's writing rules, such as a missing required bracket.

A logic error allows execution but produces an incorrect result, such as adding length and width to calculate area.

Syntax errors are fixed by correcting form; logic errors require correcting the method.

S080. A programmer calculates a rectangle's area as length + width. Identify the error and correct the formula. [3 marks]

This is a logic error because the chosen operation calculates the wrong quantity.

Correct formula: area = length x width.

For length 5 and width 3, the correct area is 15 square units, whereas addition gives 8.

S081. Trace this pseudocode: total = 0; FOR i = 1 TO 3: total = total + i; END FOR. What is the final total? [3 marks]

The loop includes i = 1, 2 and 3.

The running total becomes 1, then 3, then 6.

Therefore the final total is 6.

S082. A loop intended to run 5 times uses i < 5, starting at i = 1 and increasing by 1. How many times does it run, and how can it be corrected? [3 marks]

The loop runs for i = 1, 2, 3 and 4: four times.

The condition i < 5 excludes i = 5.

Change it to i <= 5 to run five times, keeping the start value 1 and increment 1.

S083. Why should a reusable average-calculation module accept a list and its size rather than fixed marks for one class? [3 marks]

Accepting a list and its size lets the module work with different classes and different numbers of students.

It avoids hard-coded marks and repeated code.

The module should check that the size is positive before dividing by it.

S084. Name two modules that could be used in a student-result system. [3 marks]

Input module: reads student details and marks and checks valid input.

Calculation module: calculates totals, averages and grades.

An output module may then display or print the result in a clear format.

CS-10-B-02 | Long answers

Long questions | 6 marks each

L029. Write pseudocode to input five numbers, find their sum and mean, and display both. Trace it for 2, 4, 6, 8 and 10 using a table. [6 marks]

Pseudocode:

```
total = 0
FOR i = 1 TO 5
    INPUT number
    total = total + number
END FOR
mean = total / 5
OUTPUT total, mean
```

Iteration i	Input	Total after addition
1	2	2
2	4	6
3	6	12
4	8	20
5	10	30

Final sum = 30. Mean = 30/5 = 6. The total starts at zero and each input is included exactly once.

L030. Write a generalized algorithm to find the largest of n numbers. Dry-run it for [12, 7, 19, 5] and show the largest value after each step. [6 marks]

Pseudocode, assuming n >= 1:

```
INPUT n, numbers[1..n]
largest = numbers[1]
FOR i = 2 TO n
    IF numbers[i] > largest THEN
        largest = numbers[i]
    END IF
END FOR
OUTPUT largest
```

Stage	Value considered	Largest so far
Initialization	12	12
i = 2	7	12
i = 3	19	19
i = 4	5	19

Final answer: 19. Initializing with the first value also allows the algorithm to work when all input numbers are negative.

L031. Design a modular solution for a student-result system using input, calculation and output modules. Explain how abstraction and reuse help the design. [6 marks]

Input module: reads student details, subject marks and maximum marks, and checks validity.

Calculation module: calculates total marks, percentage and grade using the required grading rules.

Output module: displays or prints the student result with clear labels.

```
studentData = ReadAndValidateInput()
result = CalculateResult(studentData)
DisplayResult(result)
```

Abstraction: the main algorithm uses each module's purpose without needing its internal details.

Reuse: the same calculation module can process different students because it accepts data as input.

Benefits: separate modules are easier to understand, check and update, and the same logic need not be copied repeatedly.

L032. Trace the algorithm: x = 1; total = 0; WHILE x <= 4: total = total + x*x; x = x + 1; END WHILE; OUTPUT total. Record x and total during each iteration and state the output. [6 marks]

The algorithm adds the squares of 1, 2, 3 and 4.

Initial values: x = 1, total = 0.

Iteration	x before addition	x*x	Total after addition	x after increment
1	1	1	1	2
2	2	4	5	3
3	3	9	14	4
4	4	16	30	5

When x becomes 5, x <= 4 is false, so the loop stops.

Output = 30, since 1 + 4 + 9 + 16 = 30.

L033. An algorithm is: total = 0; count = 1; WHILE count <= 5: INPUT mark; total = mark; END WHILE; mean = total / 4; OUTPUT mean. Identify three logic errors, rewrite it correctly and trace it for marks 10, 20, 30, 40, 50. [6 marks]

Three errors: total = mark overwrites the sum; count is never increased, causing a non-terminating loop; division by 4 is wrong for five marks.

Corrected algorithm:

```
total = 0
count = 1
WHILE count <= 5
    INPUT mark
    total = total + mark
    count = count + 1
END WHILE
mean = total / 5
OUTPUT mean
```

Mark read	Running total	Count after increment
10	10	2
20	30	3
30	60	4
40	100	5
50	150	6

The loop stops at count = 6. Mean = 150/5 = 30.

L034. Write pseudocode to count even and odd numbers in a list. Use a trace table for [3, 8, 10, 7, 5]. [6 marks]

Pseudocode, using integer inputs:

```
even = 0
odd = 0
FOR each value IN numbers
    IF value MOD 2 = 0 THEN
        even = even + 1
    ELSE
        odd = odd + 1
    END IF
END FOR
OUTPUT even, odd
```

Value	Remainder on division by 2	Even count	Odd count
3	1	0	1
8	0	1	1
10	0	2	1
7	1	2	2
5	1	2	3

Final output: 2 even numbers and 3 odd numbers. Each input increases exactly one counter.

CS-10-C-01
Front-end and back-end development

Theory scope: Conceptual roles of browser-side and server-side development only; no practical HTML/CSS or JavaScript implementation tasks.

Short questions | 3 marks each

S085. Differentiate between front-end and back-end development. [3 marks]

Front end	Back end
The part users see and interact with in the browser.	The server-side part that processes requests and manages data.
Uses HTML, CSS and browser JavaScript.	May use Python, PHP or server-side JavaScript.

Example: an order page is front end; saving the order is back end.

S086. Name three front-end technologies and state the purpose of each. [3 marks]

HTML defines a page's structure and content, such as headings and paragraphs.

CSS controls appearance, including colours, fonts and layout.

Browser JavaScript adds behaviour, such as responding when a user clicks a button.

S087. Name two technologies commonly used for back-end development. [3 marks]

Python and PHP are two commonly used back-end technologies.

They can run on a web server to process requests, check information and access databases.

Server-side JavaScript is another possible choice.

S088. Where do front-end and back-end programs run? [3 marks]

Front-end code runs in the user's browser.

Back-end code runs on a server that receives requests from the browser.

The server processes a request and returns a response, which the browser displays or uses.

S089. Give two examples of tasks handled by the back end of an online shop. [3 marks]

1. Checking a customer's login credentials and permissions.

2. Retrieving product details and prices from a database.

3. Recording orders and updating stock after a purchase.

S090. Give one example of a website feature that can work entirely in the browser. Explain why it may not need a back end. [3 marks]

A simple calculator can run entirely in the browser.

Browser JavaScript performs the calculations, while HTML and CSS provide the display.

A back end is not necessary if it does not need server-held data, user accounts or saved calculations.

S091. Why does a website with user accounts usually require a back end? [3 marks]

User accounts usually need centrally stored information and protected checks.

The back end verifies credentials and controls access to private data.

It also allows the same account to be used from different devices without trusting browser-side code alone.

S092. Describe the roles of the front end and back end when a user logs in. [3 marks]

The front end displays the login page and sends the entered details securely to the server.

The back end checks the credentials and creates an authenticated session if valid.

It returns a result that the front end uses to show success or an error.

S093. Can JavaScript be used on both the front end and back end? Explain briefly. [3 marks]

Yes. JavaScript runs in browsers for front-end interactivity.

It can also run on servers through environments such as Node.js for back-end processing.

The role depends on where the code executes and what task it performs.

S094. Why should important checks, such as permission to access an account, be enforced on the server? [3 marks]

Users can inspect or alter browser-side code and send their own requests.

Therefore server-side checks must verify identity, permissions and important data.

This prevents a user from gaining access simply by bypassing a front-end restriction.

CS-10-C-01 | Long answers

Long questions | 6 marks each

L035. Compare front-end and back-end development in terms of purpose, execution location, technologies and tasks. Explain how they work together in an online shop. [6 marks]

Aspect	Front end	Back end
Purpose	Creates the interface and interaction.	Processes requests and applies business rules.
Location	User's browser.	Web server.
Technologies	HTML, CSS and browser JavaScript.	Python, PHP or server-side JavaScript.
Data role	Displays data and collects input.	Reads, validates and stores data.

Online shop example: the front end shows products and an Add to Cart button. When a customer orders, it sends a request to the back end. The server checks stock and permissions, records the order in a database and sends a confirmation. The browser displays the result.

L036. Describe the roles of HTML, CSS and browser JavaScript in the front end. Explain how a back-end program and database support a student-result website. [6 marks]

HTML structures the student-result page with headings, labels and result information.

CSS controls its appearance, such as readable fonts, spacing and a clear layout.

Browser JavaScript responds to user actions and displays information received from the server.

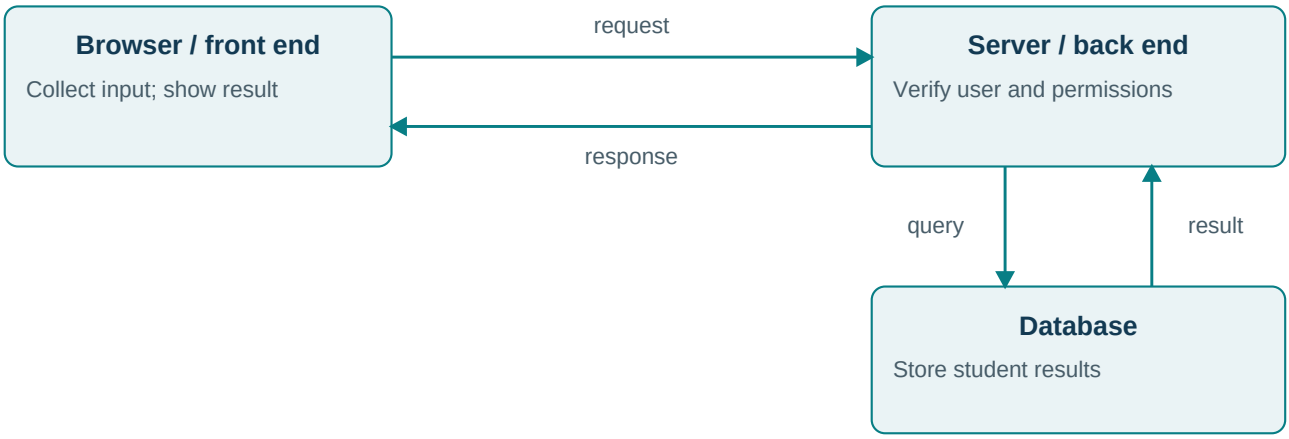
The back-end program authenticates the student and checks permission to view results. It requests the correct marks from the database.

The database stores student identities, subjects and marks in an organized form.

The server returns only authorized results to the browser. Thus the front end presents the information while the back end protects and retrieves it. No practical page code is required for this conceptual question.

L037. A user logs into a school portal to view results. Explain the request and response process from the browser to the server and back. Draw a labelled conceptual diagram. [6 marks]

1. The browser displays the school portal's login interface.
2. The user submits credentials through an encrypted connection.
3. The server verifies the credentials and establishes an authenticated session if they are correct.
4. When results are requested, the server checks that the user has permission to see that student's record.
5. The server retrieves the authorized result from the database and returns a response.
6. The front end displays the result, or an error if access is denied.



The browser presents the interface; the server performs trusted checks; the database stores the records.

L038. Compare a browser-only calculator with a website that stores users' orders. Explain why their back-end requirements differ. [6 marks]

Browser-only calculator: the browser receives numbers, performs calculations using JavaScript and shows the result. It can work without sending data to a server because all necessary logic and data are local.

Order-storage website: the browser collects product choices, but the server must check stock, determine trusted prices, verify the user and save the order in a database.

The saved order must remain available across sessions and devices and be protected from unauthorized changes.

Therefore a simple local calculator may need only a front end, whereas a shared ordering service needs a back end for reliable storage, permissions and business rules.

CS-10-D-01
Data science, AI and machine learning

Theory scope: Definitions and relationships; supervised, unsupervised and reinforcement learning; classification, regression, churn and segmentation.

Short questions | 3 marks each

S095. Define data science and state one use. [3 marks]

Data science is the use of data, analytical methods and computing to find useful information and support decisions.

It involves collecting, preparing, analysing and interpreting data.

Example: studying past sales to predict demand for the next month.

S096. Define Artificial Intelligence and give one example. [3 marks]

Artificial Intelligence (AI) enables computer systems to perform tasks that normally require human intelligence.

These tasks include recognizing language, images or patterns and making decisions.

Example: an email filter identifying likely spam.

S097. Define Machine Learning and explain its relationship with AI. [3 marks]

Machine Learning (ML) is a branch of AI in which systems learn patterns from data.

The learned patterns are used to make predictions or decisions on new data.

Thus ML is one way to build AI, but not every AI method uses ML.

S098. State two differences between data science and Machine Learning. [3 marks]

Data science is a wider process of extracting knowledge from data, including collection, cleaning and communication.

ML focuses on learning models or patterns from data.

Data scientists may use ML, but they can also use statistics and visualization without ML.

S099. Differentiate between supervised and unsupervised learning. [3 marks]

Supervised learning	Unsupervised learning
Learns from labelled examples with known target outputs.	Learns from data without supplied target labels.
Predicts an output for new cases.	Discovers groups or hidden structure.
Example: spam classification.	Example: grouping customers by buying habits.

S100. What is labelled data? Give an example. [3 marks]

Labelled data contains input examples together with the correct target output.

Example: emails marked "spam" or "not spam."

A supervised model uses these labels to learn the relationship between input features and the required output.

S101. Differentiate between classification and regression. [3 marks]

Classification predicts a category, such as spam or not spam.

Regression predicts a numerical quantity, such as next month's sales.

The main difference is whether the target is a class label or a numerical value.

S102. Is predicting next month's sales a classification or regression problem? Give a reason. [3 marks]

It is a regression problem because sales are predicted as a numerical amount.

The model can learn from past sales and other useful information.

For example, it may predict sales of 1,200 units next month rather than choose a category.

S103. Is dividing customers into groups using buying habits supervised or unsupervised learning? Explain. [3 marks]

It is usually unsupervised learning, specifically clustering.

The system groups customers with similar buying habits without being given correct group labels.

The discovered groups can help a business understand different customer needs.

S104. What is reinforcement learning? Give one simple example. [3 marks]

Reinforcement learning trains an agent through actions and feedback in the form of rewards or penalties.

The agent learns which actions improve its long-term reward.

Example: a game-playing agent learns useful moves through repeated play.

S105. How can Machine Learning help detect fraudulent transactions? [3 marks]

ML learns patterns from transaction data and can identify transactions that resemble past fraud or appear unusual.

For example, it may flag a large payment from an unfamiliar location.

The alert helps staff investigate quickly; a flag does not by itself prove fraud.

S106. What is churn prediction? How can a business use its results? [3 marks]

Churn prediction estimates which customers are likely to stop using a service.

A business can identify high-risk customers and investigate their difficulties.

It can then offer relevant support or improved service to encourage them to stay.

S107. What is behavioural segmentation? Give one benefit. [3 marks]

Behavioural segmentation groups customers by actions such as purchase frequency, spending or product preferences.

It reveals groups with similar needs.

For example, a shop can send frequent buyers and occasional buyers different, relevant offers.

S108. State two benefits of automation and scalability in Machine Learning. [3 marks]

Automation allows a trained model to perform repeated predictions quickly, reducing routine manual work.

Scalability allows the system to handle many cases, such as thousands of transactions.

People still need to monitor accuracy, fairness and unusual results.

S109. A model labels customers as high or low risk of leaving. What does a high-risk prediction mean, and what should the business do? [3 marks]

High risk means the model estimates that the customer has a greater chance of leaving, not that leaving is certain.

The business should review the customer's needs and problems.

It can offer appropriate support and later check whether the action helped.

CS-10-D-01 | Long answers

Long questions | 6 marks each

L039. Define data science, AI and Machine Learning. Compare their purposes and explain how they overlap using one real-world example. [6 marks]

Data science extracts knowledge from data through collection, cleaning, analysis, visualization and interpretation.

AI creates systems that perform tasks associated with human intelligence, such as language understanding and decision-making.

ML is a branch of AI that learns patterns from data instead of requiring a separately written rule for every case.

Overlap example: a bank collects and cleans transaction data through a data science process. An ML model learns patterns associated with fraud. The bank uses the model inside an AI-supported fraud-detection system.

Data science communicates the findings and evaluates their usefulness. AI can use methods other than ML, and data science does not always require AI.

L040. Compare supervised, unsupervised and reinforcement learning. Explain each with a suitable example. [6 marks]

Type	Learning information	Main purpose	Example
Supervised	Inputs with known target labels or values.	Predict a target for new data.	Classifying email as spam or genuine.
Unsupervised	Inputs without supplied target labels.	Find groups or structure.	Grouping customers by purchasing behaviour.
Reinforcement	Feedback through rewards or penalties after actions.	Learn a sequence of useful actions.	Learning to play a game.

Supervised learning is suitable when correct example answers are available. Unsupervised learning explores patterns without those answers. Reinforcement learning concerns an agent interacting with an environment to improve long-term reward.

L041. Explain classification, regression and clustering. Choose a suitable type for spam detection, sales prediction and customer grouping, giving reasons. [6 marks]

Classification assigns a category. Spam detection is classification because an email is assigned a class such as spam or not spam.

Regression predicts a numerical value. Sales prediction is regression because the output is a quantity, such as revenue or units sold.

Clustering discovers groups of similar observations without predefined target labels. Customer grouping is clustering when groups are formed from buying habits.

The choice follows the output required: a label, a number or discovered groups. Classification and regression are commonly supervised tasks; clustering is an unsupervised task.

L042. Explain how a company can use churn prediction and behavioural segmentation. Describe how the results can guide different actions for customers. [6 marks]

Churn prediction identifies customers likely to leave. A company can use past service use, complaints and payment information to estimate this risk.

Behavioural segmentation groups customers by similar actions, such as frequent purchases, low usage or preference for certain products.

Combined use: identify high-risk customers, then study which behavioural group each belongs to. A low-usage group may need guidance; a group with repeated service complaints may need problems resolved.

Offers should match the customer's needs rather than be sent to everyone.

The company should monitor whether actions improve retention. Predictions indicate risk and should support, rather than replace, informed human judgment.

L043. Describe the data science life cycle from collecting data to interpreting model outcomes. Use a fraud-detection example. [6 marks]

1. Collect: obtain appropriate historical transaction data and known fraud outcomes where available.

2. Prepare: remove duplicates, handle missing values and use consistent formats.

3. Analyse: study transaction amounts, times and patterns to understand the data.

4. Build: train a suitable model, such as a supervised classifier when labelled examples are available.

5. Evaluate: test it on separate data and inspect missed fraud and false alarms.

6. Interpret and communicate: explain which transactions are flagged and how investigators should use the results. Visualizations can show trends and unusual activity.

A fraud alert is evidence for investigation, not proof that a customer committed fraud.

CS-10-D-02
Data visualization and relational databases

Theory scope: Charts; centre and spread; outliers; plot interpretation; relational models, keys, schema and written database queries.

Short questions | 3 marks each

S110. State two benefits of data visualization. [3 marks]

1. Visualization summarizes many values in a form that is easier to understand.
2. It reveals trends, comparisons, relationships and unusual observations.
3. It helps analysts communicate findings and make informed decisions.

S111. Differentiate between a bar chart and a histogram. [3 marks]

A bar chart compares separate categories and normally has gaps between bars.

A histogram shows frequencies for numerical intervals; adjoining intervals normally have touching bars.

Example: subjects use categories; grouped student heights use intervals.

S112. When is a scatter plot useful? State what its two axes represent. [3 marks]

A scatter plot shows the relationship between two numerical variables.

The horizontal axis represents one variable and the vertical axis the other.

Each point represents a paired observation, such as one student's study hours and examination marks.

S113. What does a boxplot show? Name two features visible in it. [3 marks]

A boxplot summarizes the distribution using the median and quartiles.

The box extends from the first quartile to the third quartile; whiskers show further spread according to the chosen convention.

It helps compare centre and spread and may identify outliers separately.

S114. What is a dotplot? When is it useful? [3 marks]

A dotplot places dots above values on a number line, with repeated values stacked.

Each dot represents an observation.

It is useful for a small numerical dataset because individual values, frequency and gaps remain visible.

S115. Why are side-by-side bar charts useful for comparing groups? [3 marks]

Side-by-side bars place groups next to each other within each category.

They allow direct comparison, such as boys' and girls' enrolment in each subject.

A clear legend and common scale make the group differences easier to interpret.

S116. Calculate the mean and median of 2, 4, 4, 6 and 9. [3 marks]

Sum = 2 + 4 + 4 + 6 + 9 = 25; number of values = 5.

Mean = 25/5 = 5.

The ordered middle value is 4, so median = 4.

S117. Find the range of 12, 15, 18, 20 and 25. What does it describe? [3 marks]

Range = maximum - minimum = 25 - 12 = 13.

It shows the distance between the smallest and largest observation.

It is a simple measure of spread but does not describe how values are distributed between the extremes.

S118. What is an outlier? How can it affect the mean? [3 marks]

An outlier is an observation unusually far from most other values.

A very large outlier can pull the mean upward; a very small one can pull it downward.

The median is usually less affected because it depends on ordered position.

S119. What is meant by the shape, centre and spread of a dataset? [3 marks]

Shape describes the overall pattern, such as symmetry or skewness.

Centre describes a typical value, such as the mean or median.

Spread describes how variable the values are, using measures such as the range or interquartile range.

S120. A school's pass percentages are 70, 75, 80 and 85 over four years. Describe the trend and choose a suitable chart. [3 marks]

The pass percentage rises from 70% to 85%, increasing by 5 percentage points each year.

A line graph is suitable because the values are ordered over time.

Place years on the horizontal axis and pass percentage on the vertical axis.

S121. A scatter plot shows higher sales on hotter days. Why does this alone not prove that temperature is the only cause? [3 marks]

The plot shows an association between temperature and sales.

Other factors, such as season, holidays or changes in the number of customers, may also affect sales.

Therefore the association alone does not establish that temperature is the only cause.

S122. What is a relational database? Define an entity and an attribute. [3 marks]

A relational database stores related data in tables linked through keys.

An entity is a thing about which data is stored, such as a Student.

An attribute is a property of the entity, such as StudentID or Name.

S123. Differentiate between a primary key and a foreign key. [3 marks]

A primary key uniquely identifies each row of a table and must not be null.

A foreign key is a field or set of fields referring to a key in a related table.

Example: ClassID is the primary key in Class and a foreign key in Student.

S124. How does a database table differ from a database form or report? [3 marks]

A table stores data as rows and columns.

A form provides a convenient interface for entering or viewing records.

A report presents selected information in an organized format suitable for reading or printing.

S125. State the purpose of SELECT, CREATE TABLE and JOIN in SQL. [3 marks]

SELECT retrieves data from one or more tables.

CREATE TABLE defines a new table and its columns or constraints.

JOIN combines related rows from tables using a matching condition.

S126. How can data stored in a database be used in Machine Learning? [3 marks]

A database organizes and stores records needed for training or prediction.

Queries retrieve relevant fields, and the data is cleaned and prepared for the model.

For example, past sales records can provide features and target sales values for a prediction model.

S127. Write a SELECT query to display Name and Marks from Student where Marks >= 80. [3 marks]

```
SELECT Name, Marks
FROM Student
WHERE Marks >= 80;
```

SELECT chooses the two columns, FROM specifies the table and WHERE keeps only rows with marks of at least 80.

CS-10-D-02 | Long answers

Long questions | 6 marks each

L044. For marks [10, 12, 12, 14, 17], calculate the mean, median, mode and range. Replace 17 with 50 and explain the effect on the mean and median. [6 marks]

Original dataset: [10, 12, 12, 14, 17].

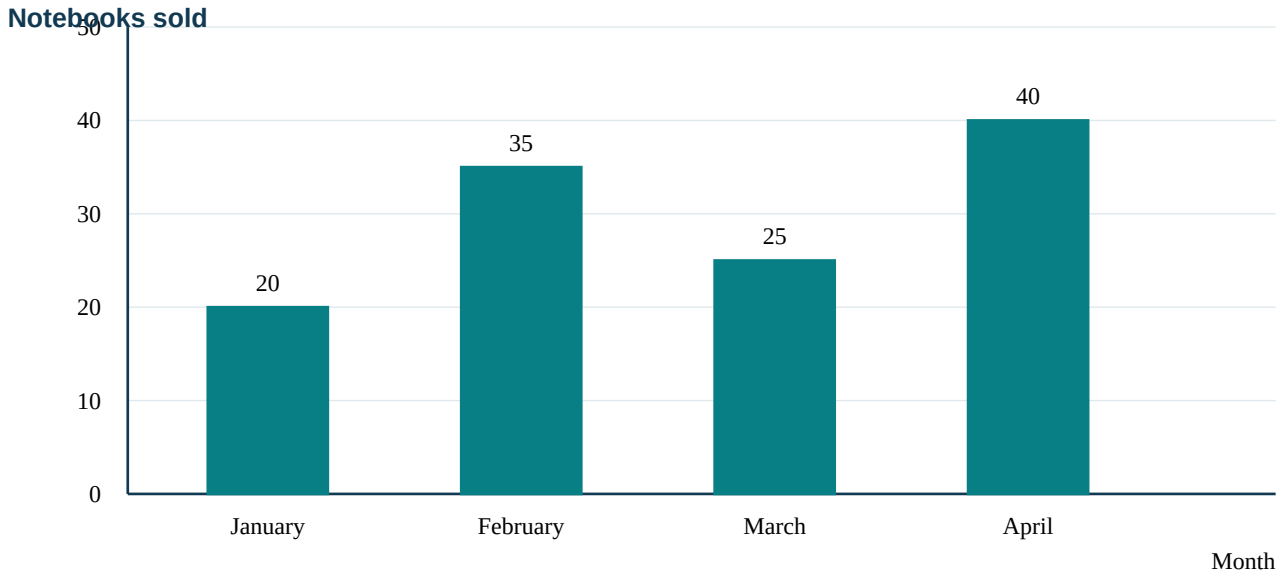
Sum = 65, so mean = 65/5 = 13. Median = 12, mode = 12 and range = 17 - 10 = 7.

After replacing 17 with 50: [10, 12, 12, 14, 50].

Sum = 98, so mean = 98/5 = 19.6. Median = 12, mode = 12 and range = 50 - 10 = 40.

The extreme value raises the mean from 13 to 19.6 and greatly increases spread. The median remains 12 because the middle position does not change. This illustrates why the median can describe a typical value better when a dataset contains an extreme observation.

L045. A shop sells 20, 35, 25 and 40 notebooks in January, February, March and April. Draw a labelled bar chart. State two observations and explain which chart would best emphasize change over time. [6 marks]



Horizontal axis: month. Vertical axis: notebooks sold, with a zero baseline and equal intervals.

Observation 1: April has the highest sales, 40 notebooks; January has the lowest, 20.

Observation 2: sales rise from January to February, fall in March and rise again in April. Overall, April exceeds January by 20 notebooks.

For emphasizing change over time, a line graph with months in chronological order is especially useful. A bar chart clearly compares the monthly totals. Both must use labels and a consistent numerical scale.

L046. Choose suitable visualizations for subject enrolments, grouped heights, study hours versus marks, and the distribution of marks including outliers. Explain each choice. [6 marks]

1. Subject enrolments: a bar chart compares counts for separate subject categories.

2. Grouped heights: a histogram shows frequencies in consecutive numerical height intervals.

3. Study hours versus marks: a scatter plot places paired numerical observations on two axes and helps reveal an association.

4. Distribution of marks including outliers: a boxplot shows median, quartiles and spread, and can mark outliers separately.

The chart must match the data type and question. Categories need comparisons; continuous groups need frequency intervals; paired measurements need a relationship plot; distribution comparisons benefit from a summary of centre and spread. Label axes and units clearly.

L047. Compare bar charts, histograms, scatter plots, dotplots and boxplots. Give one suitable use of each. [6 marks]

Display	What it shows	Suitable example
Bar chart	Values or counts for separate categories.	Enrolment in different subjects.
Histogram	Frequency across numerical intervals.	Heights grouped into 5 cm intervals.
Scatter plot	Paired numerical values and their association.	Study hours and marks.
Dotplot	Individual numerical observations and repeated values.	Marks of a small class.
Boxplot	Median, quartiles and spread, often with outliers.	Comparing marks in two classes.

Choose a plot according to the purpose. Use clear labels, units and a fair scale. A chart should communicate the data accurately and should not imply that correlation proves causation.

L048. Two classes have marks A = [40, 50, 60] and B = [10, 50, 90]. Calculate their means and ranges. Compare their centres and spreads and explain why one average does not describe all differences. [6 marks]

Class A: mean = (40+50+60)/3 = 150/3 = 50. Range = 60-40 = 20.

Class B: mean = (10+50+90)/3 = 150/3 = 50. Range = 90-10 = 80.

The two classes have the same mean and also the same median, 50. However, Class B has a much wider spread.

Class A's marks are closer to 50, whereas Class B contains a much lower and a much higher mark.

Therefore equal averages do not mean equal distributions. Centre and spread must be considered together when describing performance.

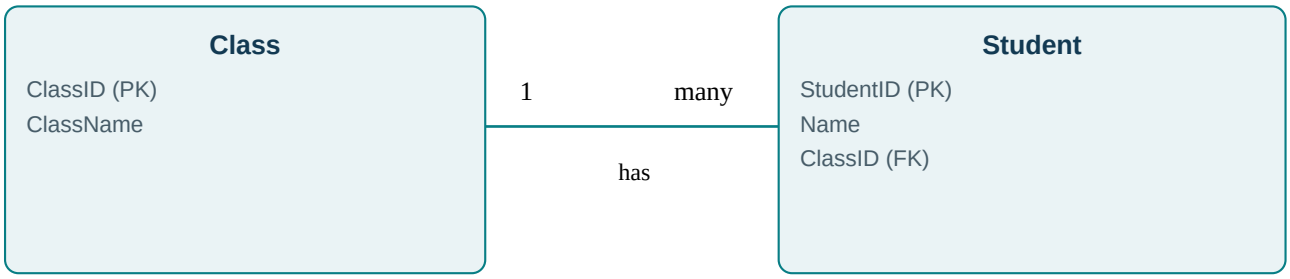
L049. A school stores Student(StudentID, Name, ClassID) and Class(ClassID, ClassName). Identify entities, attributes, primary keys and the foreign key. Draw the one-to-many E-R relationship and write the relational schema. [6 marks]

Entities: Student and Class.

Student attributes: StudentID, Name, ClassID. Class attributes: ClassID and ClassName.

Primary keys: Student.StudentID and Class.ClassID.

Foreign key: Student.ClassID references Class.ClassID.



One class can have many students, while each student record in this simplified design belongs to one class.

Relational schema:

```
Class(ClassID PK, ClassName)
Student(StudentID PK, Name,
        ClassID FK -> Class.ClassID)
```

The foreign key links each student to a valid class record and avoids storing the class name repeatedly in every student row.

L050. Using Student(StudentID, Name, ClassID) and Class(ClassID, ClassName), write a query to list student names with class names using JOIN. Explain the joining field and the purpose of a primary key. [6 marks]

SQL query:

```
SELECT s.Name, c.ClassName
FROM Student AS s
JOIN Class AS c
    ON s.ClassID = c.ClassID;
```

The joining field is ClassID. Each student's ClassID is matched with the same ClassID in the Class table.

The primary key Class.ClassID uniquely identifies a class. Student.StudentID uniquely identifies a student.

The foreign key Student.ClassID records the student's class relationship.

The query retrieves the student name from Student and the matching class name from Class. An inner JOIN returns rows with a matching class. The aliases s and c make the column references shorter and unambiguous.

L051. Write SQL to create Student with StudentID as an integer primary key, Name as text and Marks as an integer. Then write a query to show names and marks for students scoring at least 80, ordered by marks from highest to lowest. [6 marks]

```
CREATE TABLE Student (
    StudentID INTEGER PRIMARY KEY,
    Name VARCHAR(100),
    Marks INTEGER
);

SELECT Name, Marks
FROM Student
WHERE Marks >= 80
ORDER BY Marks DESC;
```

CREATE TABLE defines the table. INTEGER is used for student IDs and marks; VARCHAR stores text names. PRIMARY KEY requires a unique, non-null StudentID.

The SELECT query retrieves names and marks. WHERE includes marks of 80 or more. ORDER BY Marks DESC arranges the result from highest to lowest marks.

This is standard illustrative SQL; some database tools use different text-type names.

L052. Explain logical and physical database design. For a library with Book and Member entities, add a Loan entity to record borrowing, identify suitable keys and explain the relationships. [6 marks]

Logical design identifies entities, attributes, keys and relationships without focusing on storage details. Physical design implements the design using actual tables, data types, indexes and storage arrangements in a database system.

Example schema:

```
Book(BookID PK, Title)
Member(MemberID PK, Name)
Loan(LoanID PK, BookID FK, MemberID FK,
     BorrowDate, ReturnDate)
```

Loan.BookID references Book.BookID; Loan.MemberID references Member.MemberID.



A book can appear in many loan records over time, and a member can have many loans. Each loan records one book being borrowed by one member. Here Book represents an individually identified copy. The Loan entity preserves borrowing history.

L053. Monthly sales are 100, 120, 140 and 160 units. Advertising also rises during these months. Describe the trend, suggest a useful plot and explain why seasonal demand could be a confounding factor. [6 marks]

Trend: sales increase by 20 units each month, from 100 to 160 units.

Useful plot: a line graph with month on the horizontal axis and units sold on the vertical axis shows the sales trend clearly. A scatter plot of advertising spending against sales can examine their association if paired spending values are available.

Confounding factor: seasonal demand might raise sales during the same months that advertising rises. More advertising and more sales could both occur during a busy season.

Therefore the data alone does not prove that advertising caused the increase. Information about season, prices, customer numbers and other changes would help interpretation.

CS-10-D-03
Applying the data science life cycle

Theory scope: Problem formulation; data sources, collection, storage and preparation; model choice; interpreting results; mean squared error.

Short questions | 3 marks each

S128. State the main stages of the data science life cycle. [3 marks]

1. Understand and define the problem.

2. Gather, store, clean and prepare relevant data.

3. Explore the data, build a suitable model, evaluate it, and interpret and communicate the results.

The stages may be repeated when new findings reveal problems.

S129. Turn the business question "What will our sales be next month?" into a data science problem. [3 marks]

Formulate it as: "Use historical sales and relevant factors to predict the numerical sales total for next month."

This is a supervised regression problem when past input-output examples are available.

The predicted amount supports stock and staffing decisions.

S130. What data could help predict a store's sales? Give two examples. [3 marks]

Past daily or monthly sales show earlier demand patterns.

Prices, promotions, holidays or seasonal information may help explain changes.

These data should correspond to the same dates and use consistent units.

S131. State two sources of data for a data science project. [3 marks]

One source is an internal database, such as a store's transaction records.

Another source is a survey collecting relevant customer information.

Suitable public datasets or sensor records may also be used if their quality and permitted use are checked.

S132. Why must collected data be cleaned before modelling? [3 marks]

Raw data can contain duplicates, missing values and incorrect entries.

Cleaning reduces errors that could mislead the model.

For example, the same sale recorded twice would exaggerate demand unless the duplicate is removed.

S133. Give two examples of formatting or preparing data for analysis. [3 marks]

Convert dates into one consistent format so records can be correctly ordered and matched.

Convert numerical values into consistent units or types, such as all prices in rupees.

Also handle missing entries and check invalid values before modelling.

S134. What is model building in the data science life cycle? [3 marks]

Model building means choosing a suitable method and using prepared data to learn a relationship or pattern.

The model can then predict or classify new cases.

Example: fitting a sales-prediction model using historical records.

S135. Name a problem suitable for simple linear regression. [3 marks]

Simple linear regression models a numerical outcome using one predictor.

Example: estimating sales from advertising spending when a straight-line relationship is a reasonable approximation.

The prediction is numerical, and the observed relationship should be checked before use.

S136. What is a decision tree? State one possible use. [3 marks]

A decision tree makes predictions through a sequence of feature-based questions or rules.

Each path leads to an output or prediction.

Example: classifying customers as likely or unlikely to leave using usage and complaint information.

S137. State two factors to consider when selecting a model. [3 marks]

1. The task: choose a method suitable for a numerical prediction, category prediction or grouping problem.

2. The data: consider quality, quantity and whether labels are available.

3. Also consider interpretability and performance on suitable evaluation data.

S138. What is the purpose of evaluating a predictive model? [3 marks]

Evaluation checks how well the model performs on data not used to fit it.

It identifies prediction errors and helps compare candidate models.

A model should be useful for the original problem rather than merely fit its training examples well.

S139. Write the formula for Mean Squared Error (MSE). What does a smaller MSE indicate when comparing the same prediction task? [3 marks]

MSE = [sum of (actual value - predicted value)^2] / n, where n is the number of predictions.

It is the average squared prediction error.

For the same task, data and units, a smaller MSE indicates smaller average squared errors.

S140. Calculate MSE when actual values are [2, 4] and predicted values are [3, 3]. [3 marks]

Errors: 2-3 = -1 and 4-3 = 1.

Squared errors: 1 and 1; sum = 2.

MSE = 2/2 = 1.

S141. Why should predictions be interpreted in the context of the original business problem? [3 marks]

A prediction becomes useful when it is connected to the original decision.

For example, predicted sales can guide stock purchases, while churn risk can guide customer support.

Interpretation should consider uncertainty and limitations rather than treat every prediction as certain.

CS-10-D-03 | Long answers

Long questions | 6 marks each

L054. A shop wants to predict next year's sales from the previous three years. Explain problem definition, data collection, storage, preparation, model building and evaluation. [6 marks]

1. Define the problem: predict next year's sales to support planning. State whether the target is total sales or monthly sales.

2. Collect: obtain the last three years' sales with dates, prices and relevant seasonal or promotional information.

3. Store: keep records in an organized database or structured file with consistent fields.

4. Prepare: correct errors, remove duplicates, handle missing values and use consistent dates and units.

5. Model: choose an appropriate prediction method, such as regression, and fit it using suitable earlier data.

6. Evaluate and interpret: test on separate later data, calculate an error measure such as MSE, and explain how reliable the forecast is. Use it to guide stock or staffing decisions.

L055. A company says, "Our customers are leaving us." Formulate a data science problem, suggest useful data, choose a suitable learning approach and explain how predictions can guide action. [6 marks]

Problem: predict which customers are likely to stop using the company's service during a defined future period.

Data: historical usage, complaints, payments, length of membership and known past leaving outcomes.

Approach: supervised classification if previous customers are labelled as left or stayed. A decision tree is one possible model.

Process: clean the data, train the model on suitable records and evaluate it on separate data.

Action: contact high-risk customers, investigate their difficulties and offer relevant support or service improvements.

Interpretation: risk is not certainty. The company should monitor false predictions and whether the chosen actions actually improve retention.

L056. Calculate MSE for actual values [3, 5, 7, 9] and predicted values [2, 6, 6, 10]. Show errors, squared errors and the final average in a table. Explain the purpose of this metric. [6 marks]

Formula: MSE = sum of squared errors / n; error = actual - predicted.

Actual	Predicted	Error	Squared error
3	2	1	1
5	6	-1	1
7	6	1	1
9	10	-1	1

Sum of squared errors = 1+1+1+1 = 4.

Number of predictions = 4.

MSE = 4/4 = 1.

MSE measures the average squared difference between predictions and actual values. Squaring makes all contributions non-negative and gives greater weight to larger errors. A lower MSE on the same evaluation task indicates more accurate numerical predictions.

L057. Model A predicts [11, 18, 31] and Model B predicts [12, 19, 29] for actual values [10, 20, 30]. Calculate each MSE, identify the better result on these data and justify your choice. [6 marks]

Actual	Model A	A squared error	Model B	B squared error
10	11	1	12	4
20	18	4	19	1
30	31	1	29	1

Model A: sum of squared errors = 1+4+1 = 6; MSE = 6/3 = 2.

Model B: sum of squared errors = 4+1+1 = 6; MSE = 6/3 = 2.

Conclusion: neither is better according to MSE on these data; they tie. Their individual errors differ, but their average squared error is equal. Other valid evidence would be needed to distinguish their performance.

L058. Explain how to gather data from a database, remove duplicate records, handle missing values and format the data for modelling. Use a simple school or shop example. [6 marks]

1. Define what data is needed, such as dates, quantities and prices for a shop's sales project.

2. Query the database for those fields and the relevant time period. Avoid collecting unnecessary personal information.

3. Store the extracted records with clear column names and a record identifier where appropriate.

4. Remove genuine duplicate records carefully so one sale is not counted twice.

5. Handle missing values by investigating them and applying a justified approach, such as correcting the source or excluding unusable records.

6. Standardize dates, data types and units, then check values for obvious errors. The prepared data can then be used for analysis and model building.

L059. Compare simple linear regression and a decision tree at an introductory level. Suggest a suitable model for predicting sales and one for classifying customers as likely or unlikely to leave. Give reasons. [6 marks]

Simple linear regression models a numerical target as a straight-line function of one predictor, for example sales predicted from advertising spending.

A decision tree divides data through a sequence of feature-based questions and produces an output at each final branch. It can be used for classification or regression.

For predicting a numerical sales amount, simple linear regression may be suitable if one useful predictor has an approximately linear relationship with sales.

For classifying customers as likely or unlikely to leave, a classification decision tree is suitable when labelled examples are available.

The final choice should be checked against the available data and evaluation results.

CS-10-E-01
IoT, Blockchain and cloud computing

Theory scope: Foundational IoT components; applications and scenario suitability; Blockchain uses; cloud benefits.

Short questions | 3 marks each

S142. Define the Internet of Things (IoT). Give one example. [3 marks]

The Internet of Things connects physical devices that collect and exchange data through a network.

Devices can use the information to monitor conditions or trigger actions.

Example: a soil-moisture system that reports readings and controls irrigation.

S143. Name three foundational components of an IoT system. [3 marks]

1. Sensors or devices collect information from the physical environment.
2. Connectivity transfers the information through a network.
3. Data processing interprets the readings and may trigger an action, such as switching on a pump.

S144. What is the role of a sensor in IoT? [3 marks]

A sensor measures a physical condition, such as temperature, movement or soil moisture.

It converts the measurement into data that a controller can use.

This provides the input an IoT system needs for monitoring and automatic decisions.

S145. How do connectivity and data processing support IoT? [3 marks]

Connectivity carries readings between devices, controllers or cloud services.

Data processing analyses those readings according to rules or models.

For example, a controller receives a high temperature and activates an appropriate cooling response.

S146. Explain how a soil-moisture sensor can help a farmer. [3 marks]

A soil-moisture sensor measures how wet or dry the soil is.

A connected controller can use the reading to switch irrigation on when needed and off when moisture is sufficient.

This can reduce wasted water and unnecessary manual checking.

S147. A thermometer displays a temperature but cannot communicate it. Is it automatically an IoT device? Give a reason. [3 marks]

No. A digital display alone does not make a thermometer an IoT device.

IoT requires network connectivity for exchanging data with another device or service.

The thermometer would become part of IoT if its readings were communicated through such a connection.

S148. Define Blockchain. State one application. [3 marks]

Blockchain is a shared digital ledger that records data in blocks linked using cryptographic hashes.

Participants follow agreed rules to validate updates.

Example: recording cryptocurrency transactions or tracking goods through a supply chain.

S149. How does linking blocks with hashes help detect changes? [3 marks]

A block includes a hash reference linking it to an earlier block.

Changing earlier data changes its hash, so the link no longer matches the expected record.

This makes unauthorized alterations detectable; network validation further helps protect the shared history.

S150. Give two applications of Blockchain other than cryptocurrency. [3 marks]

Supply-chain tracking records the movement of products between participants.

Asset administration records ownership or transfers of assets.

In both cases, a shared, tamper-evident history can help participants check records and reduce disputes.

S151. What is a shared or distributed ledger? [3 marks]

A distributed ledger is a record maintained across multiple participants or computers rather than only one central copy.

Participants follow rules to agree on valid updates.

This can improve shared verification, but the level of access depends on the system's permissions.

S152. State two benefits of cloud computing. [3 marks]

Cloud computing provides computing resources through a network.

Benefits include access from different locations and the ability to increase storage or processing capacity as needs grow.

Shared services may also reduce the need to maintain all infrastructure locally.

S153. How can cloud computing support an IoT system? [3 marks]

The cloud can receive and store readings from many connected devices.

It can provide processing, dashboards and alerts that users access remotely.

Example: a farm dashboard stores moisture readings and shows trends from several fields.

S154. Why can Blockchain not guarantee that a faulty sensor's original reading is true? [3 marks]

Blockchain can help show whether a recorded value was changed after recording.

It cannot independently prove that the sensor measured reality correctly.

Therefore a false reading from a faulty or compromised sensor can still be recorded; reliable sensors and validation are also needed.

CS-10-E-01 | Long answers

Long questions | 6 marks each

L060. Explain how IoT works through sensors, connectivity and data processing. Use smart irrigation as an example and draw a labelled diagram. [6 marks]

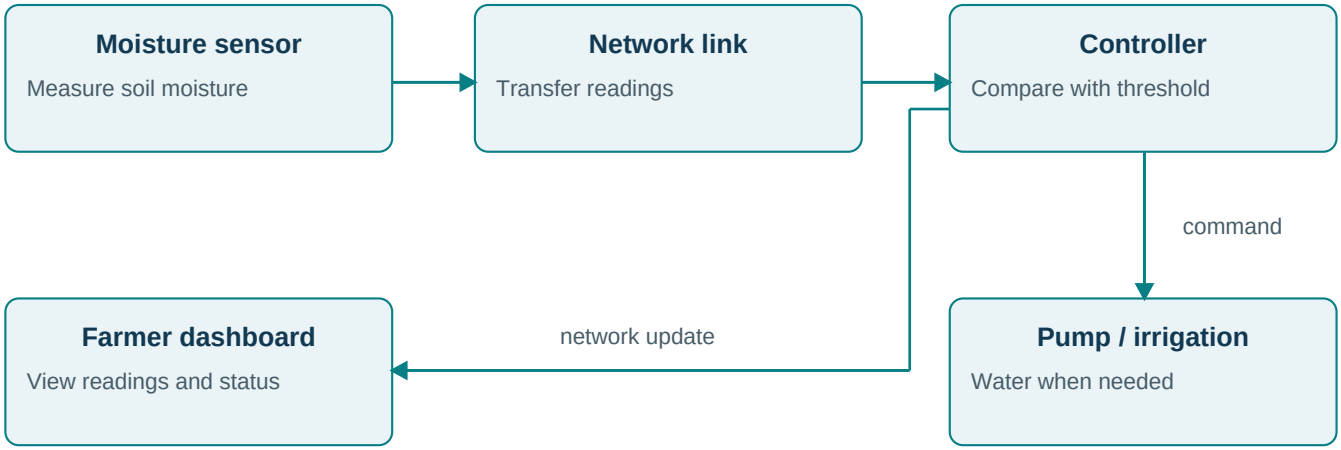
Sensors: a soil-moisture sensor measures water content in the soil.

Connectivity: the reading is sent to a controller or service through a network connection.

Processing: the controller compares the reading with a suitable moisture threshold.

Action: if the soil is too dry, the system activates an irrigation pump; when enough moisture is reached, it stops watering.

Monitoring: readings and actions can be sent to a farmer's dashboard for checking.



This reduces repeated manual checks and can save water. Reliable thresholds, safe equipment and working communication are still needed for correct operation.

L061. Explain the use of IoT in healthcare, farming, homes, transport and industry. [6 marks]

1. Healthcare: connected monitoring devices send patient readings to staff, supporting timely observation and alerts.

2. Farming: soil-moisture sensors help control irrigation and reduce unnecessary watering.

3. Homes: connected lights, thermostats and security sensors allow monitoring and control.

4. Transport: vehicle-location devices help track fleets and improve route planning.

5. Industry: machine sensors monitor temperature, vibration or operating conditions to identify possible problems.

6. Common process: sensors collect data, networks carry it and software processes it to support a useful action or decision.

The benefit is more timely information and control, provided the system is maintained and secured.

L062. Describe how Blockchain records and validates transactions. Explain its use in cryptocurrency and tracking assets. [6 marks]

1. A participant proposes a transaction, often authorizing it with a digital signature.

2. The network checks the transaction under its agreed validation rules.

3. Valid transactions are recorded in a block.

4. Cryptographic hashes link blocks, and participants maintain the agreed ledger history.

Cryptocurrency use: the ledger records transfers and helps the network verify ownership and prevent spending the same funds twice.

Asset tracking: a system can record a product's movement or changes in ownership, creating an auditable history for authorized participants.

Blockchain makes changes to recorded history harder to hide; it does not automatically establish the truth of every real-world claim entered into it.

L063. Explain how IoT and Blockchain can work together to record medicine temperatures during transport. Describe the benefit and one limitation of this arrangement. [6 marks]

1. Sensors measure medicine temperatures during transport.

2. A connected gateway collects readings and submits authorized records, or hashes of records, to the Blockchain system.

3. The network validates updates and stores an agreed, time-ordered history.

4. Linked records make later unauthorized changes easier to detect.

5. Authorized staff can review temperatures at different stages and investigate any readings outside the permitted range.

Benefit: transport partners share a traceable record and can detect changes to recorded information.

Limitation: a faulty sensor may submit a false value. Blockchain cannot correct the original measurement, so device security, calibration and suitable data checks remain necessary.

L064. A school needs shared storage, access from home and room to store more data later. Explain five benefits of using cloud computing for these needs. [6 marks]

1. Remote access: authorized teachers and students can open shared files from school or home.

2. Shared storage: one organized location makes resources easier to find and distribute.

3. Collaboration: users can work on shared documents and see current versions.

4. Scalability: storage can be expanded as the school creates more files.

5. Maintenance support: the provider manages much of the hosted infrastructure and service updates.

6. Recovery support: suitable backup and version-history features can help restore work, depending on the service selected.

The school must configure access permissions and account security. The benefits depend on reliable connectivity and the chosen service's features.

L065. Compare an ordinary offline appliance with an IoT appliance. Explain the added roles of sensors, network connection, processing and action in a smart system. [6 marks]

An offline appliance performs its local function without exchanging information through a network. For example, an ordinary fan runs when a user operates its switch.

An IoT fan may include a temperature sensor, network connectivity and a controller.

Sensor: measures room conditions. Network: shares readings or receives authorized remote commands. Processing: compares conditions with settings. Action: adjusts fan operation when needed.

A dashboard may show device status remotely.

Thus connectivity and data processing add monitoring and coordinated control. Merely being electronic or automatic does not make an appliance IoT; it must be connected to exchange data.

CS-10-E-02
Applications of AI

Theory scope: Speech recognition; personal assistants; Quran memorization support; NLP applications; robotics.

Short questions | 3 marks each

S155. Define speech recognition and give one application. [3 marks]

Speech recognition identifies or interprets spoken language using computer processing.

For example, a voice assistant recognizes the command "Set an alarm."

Speech-to-text is an application that converts recognized speech into written words.

S156. How can a personal assistant use speech recognition? [3 marks]

The assistant captures the user's spoken request through a microphone.

Speech recognition identifies the words, and language processing helps interpret the request.

The assistant can then perform an action, such as setting a reminder or answering a question.

S157. Explain one way AI can support a Quran memorization application. [3 marks]

An application can compare a learner's recitation with an expected passage and identify possible missed or changed words.

It can give practice feedback and allow repeated attempts.

Such feedback supports learning; a qualified teacher is still valuable for checking pronunciation and recitation.

S158. Differentiate between speech recognition and speech-to-text output. [3 marks]

Speech recognition is the broader process of identifying spoken words or commands.

Speech-to-text uses recognized speech to produce written text.

For example, recognizing "Play music" may trigger an action, while dictating a paragraph produces text.

S159. How can voice-based authentication use speech information? [3 marks]

Voice-based authentication may analyse a speaker's vocal characteristics to compare them with a registered voice profile.

Recognizing who is speaking is called speaker recognition or verification; recognizing the words is speech recognition.

Secure systems may combine voice information with other checks because voices can be imitated or recorded.

S160. Define Natural Language Processing (NLP). Give one example. [3 marks]

Natural Language Processing is an AI area concerned with analysing and using human language.

It can identify meaning, patterns or intent in text and language data.

Example: translating an English sentence into Urdu.

S161. How does NLP help in email filtering? [3 marks]

NLP examines features of an email, such as words, phrases and patterns associated with spam.

A trained filter can classify messages as likely unwanted or suspicious.

This reduces inbox clutter and helps limit exposure to scams, although errors are still possible.

S162. State one use each of language translation and document analysis. [3 marks]

Language translation converts meaning from one language into another, such as English to Urdu.

Document analysis extracts useful information or summarizes content from documents.

For example, it can identify the main topics in a set of reports.

S163. What is predictive text? Give a daily-life example. [3 marks]

Predictive text suggests words or phrases based on what the user has already typed and learned language patterns.

Example: a phone suggests "morning" after "Good."

It speeds up typing, but users should check that the suggestion matches their meaning.

S164. What is sentiment analysis? What can it show about customer reviews? [3 marks]

Sentiment analysis estimates the attitude expressed in language, such as positive, negative or neutral.

Applied to customer reviews, it can summarize satisfaction or common complaints.

The results may need human checking because sarcasm and context can be difficult.

S165. Give two applications of robotics in manufacturing. [3 marks]

Industrial robots can assemble or weld product parts repeatedly.

They can also pack or move goods in a factory.

Their precise, repeatable operation can improve efficiency and reduce human exposure to dangerous tasks.

S166. How can robots help in search and rescue operations? [3 marks]

Robots can enter unstable or dangerous areas that may be unsafe for rescuers.

Cameras and sensors help locate trapped people and send information to rescue teams.

Human operators use the information to plan a safer response.

S167. State one use each of robots in homes and farming. [3 marks]

Home use: a robotic vacuum navigates rooms and removes dirt.

Farming use: a robot may monitor crops or perform tasks such as precision weeding.

Sensors and control software help each robot respond to its surroundings.

S168. How can speech recognition support security-related work? State one privacy concern. [3 marks]

Speech recognition can convert authorized audio into searchable text to help analysts review relevant recordings.

A privacy concern is that private conversations might be collected or used without appropriate authority.

Access controls, limited collection and human oversight are therefore important.

CS-10-E-02 | Long answers

Long questions | 6 marks each

L066. Describe five applications of speech recognition, including personal assistants, speech-to-text, Quran memorization support, authentication and security-related use. [6 marks]

1. Personal assistants recognize spoken commands and help users perform tasks such as setting alarms.

2. Speech-to-text applications turn dictated words into written text, supporting faster input and accessibility.

3. Quran memorization tools can compare recitation with an expected passage and provide practice feedback on possible word errors.

4. Voice-based authentication uses speaker characteristics to help verify identity; this is speaker recognition and is distinct from recognizing the spoken words.

5. Security-related systems may transcribe appropriately obtained recordings for review and search.

6. These systems should be checked for errors, accent differences, privacy and misuse. Sensitive tasks need suitable safeguards and human review.

L067. Explain the use of NLP in spam filtering, language translation, document analysis, predictive text and sentiment analysis. [6 marks]

1. Spam filtering: NLP analyses email content and patterns to identify unwanted or suspicious messages.

2. Translation: it converts language content into another language, helping people understand material across language barriers.

3. Document analysis: it extracts topics, important information or summaries from documents.

4. Predictive text: it suggests likely words or phrases based on the text already entered.

5. Sentiment analysis: it estimates whether language expresses a positive, negative or neutral attitude, such as in reviews.

6. Benefit and limitation: NLP helps process large amounts of language quickly, but context, ambiguity and sarcasm can cause mistakes. Users should review important outputs.

L068. Explain applications of robotics in manufacturing, search and rescue, home cleaning and agriculture. Describe one benefit of using robots in a dangerous task. [6 marks]

Manufacturing: robots can weld, assemble, handle materials and package products with repeatable movements.

Search and rescue: mobile robots use cameras and sensors to inspect damaged or hazardous areas and help locate people.

Home cleaning: robotic vacuums navigate rooms and collect dirt with limited manual effort.

Agriculture: robots can inspect crops, remove weeds or help with precision application of resources.

Dangerous-task benefit: a robot can enter an unstable building or hazardous industrial area while people remain at a safer distance.

Robots still need suitable sensors, programming, maintenance and human supervision. Their value depends on matching their capabilities to the environment and task.

L069. For each task, identify a suitable AI area and explain its role: typing spoken words, translating a message, classifying reviews, finding trapped people and answering a spoken question. [6 marks]

Task	Suitable AI area	Role
Typing spoken words	Speech recognition / speech-to-text.	Converts recognized speech into written text.
Translating a message	NLP.	Transfers meaning between languages.
Classifying reviews	NLP / sentiment analysis.	Estimates whether opinions are positive, negative or neutral.
Finding trapped people	Robotics with sensors and possibly computer vision.	Searches dangerous places and sends useful observations.
Answering a spoken question	Speech recognition and NLP.	Recognizes the question, interprets it and supports a response.

A single application can combine several AI areas. A voice assistant, for example, needs to recognize speech and understand the request before responding.

CS-10-E-03
Social and ethical implications of AI

Theory scope: Unfair outcomes; social injustice; responsibility of designers; ethical norms and human benefit.

Short questions | 3 marks each

S169. What is AI bias? Give one simple example. [3 marks]

AI bias is a systematic unfair tendency in an AI system's results.

Example: a speech system may work poorly for one regional accent if its training data did not represent that accent well.

This can unfairly reduce access to the service.

S170. How can unrepresentative training data produce unfair decisions? [3 marks]

Training data teaches a model which patterns to use.

If some groups are missing or poorly represented, the model may perform worse for them.

For example, a recognition system trained mainly on one accent may misinterpret speakers with another accent.

S171. State two ethical responsibilities of AI designers. [3 marks]

Designers should use suitable, representative data and test for unfair outcomes.

They should protect privacy and limit use of personal information.

They should also provide clear accountability and human review for important decisions.

S172. Why is human review important for high-impact AI decisions? [3 marks]

High-impact decisions can affect jobs, education, safety or access to services.

AI can make mistakes or reproduce biased patterns.

Human review allows context to be considered, errors to be challenged and responsibility to remain with accountable people.

S173. Give two ways AI can benefit society. [3 marks]

AI can help doctors analyse medical information and identify possible problems.

It can also improve accessibility through speech-to-text and language tools.

These benefits depend on careful design, appropriate testing and responsible use.

S174. How can improper use of AI harm a group of people? [3 marks]

An unfair system may wrongly reject a group from jobs, loans or services.

Misuse may also expose private information or spread harmful false content.

These harms can reinforce existing inequalities and reduce trust.

S175. Why should an AI system protect personal data? [3 marks]

Personal data may reveal sensitive details about a person's life.

Protecting it reduces the risk of unauthorized access, discrimination or misuse.

AI systems should collect only necessary data and use appropriate permissions, security and retention practices.

S176. What does transparency mean when an organization uses AI? [3 marks]

Transparency means clearly explaining that AI is being used, its purpose and relevant limitations.

People should receive understandable information about how it influences important decisions.

This supports informed use, accountability and the ability to question errors.

S177. A speech-recognition system works poorly for one regional accent. What fairness problem does this show, and how could it be reduced? [3 marks]

The system provides unequal quality of service to speakers of that accent.

The organization should include more representative speech data and test performance across accents.

It should also offer a way to correct errors and obtain human assistance.

CS-10-E-03 | Long answers

Long questions | 6 marks each

L070. Explain how AI can cause social injustice. Use two examples and suggest measures to make AI decisions fairer. [6 marks]

AI may produce unfair outcomes when its data, design or deployment disadvantages some groups.

Example 1: a recruitment model trained on biased past hiring records may reject suitable applicants from an underrepresented community.

Example 2: a speech system trained mainly on one accent may misunderstand other speakers and make a service harder for them to use.

Reduce harm by collecting representative data, checking performance across groups and correcting unfair patterns.

Provide human review and a way to appeal important decisions.

Monitor the system after deployment because errors and changing conditions may create new problems. Responsibility stays with the organization using the system.

L071. Describe five ethical principles for designing AI systems that benefit people. [6 marks]

1. Fairness: test whether the system treats different groups unjustly and reduce avoidable bias.

2. Privacy: collect only necessary personal data and protect it from unauthorized access and misuse.

3. Transparency: explain the system's purpose, relevant limitations and use in decisions.

4. Accountability: identify responsible people and provide a way to challenge harmful or incorrect decisions.

5. Human benefit and safety: design the system to solve useful problems while testing accuracy and possible harm.

Human oversight supports all five principles, especially where errors affect rights, safety or access to important opportunities. Ethical design must continue through monitoring and improvement after deployment.

L072. A recruitment system repeatedly rejects suitable applicants from one community. Explain possible causes, effects on people and steps the organization should take. [6 marks]

Possible causes: past recruitment records may reflect discrimination; some groups may be underrepresented; or a feature may act as an unfair proxy for community membership.

Effects: qualified applicants lose opportunities, the organization misses talent and trust is damaged.

Response: pause or restrict the harmful automated decision, investigate the data and rules, and have people review affected applications.

Improve the training data, remove unjustified factors and test results across relevant groups.

Provide explanations and an appeal process where appropriate.

Continue monitoring after changes. A system's output should not be accepted automatically merely because it is generated by a computer.

L073. Discuss responsible AI use in public services or law enforcement, referring to fairness, privacy, accuracy, human review and accountability. [6 marks]

Fairness: test whether error rates or decisions unfairly disadvantage particular groups.

Privacy: use only necessary, appropriately obtained data and protect it with access controls.

Accuracy: evaluate the system in the actual setting and communicate uncertainty; a prediction is not proof of wrongdoing.

Human review: qualified people should examine high-impact outputs before acting and consider the wider context.

Accountability: the responsible organization must keep suitable records, investigate complaints and correct errors.

Transparency: explain the purpose and limitations of the system so people can understand and challenge relevant decisions.

These measures help AI support public benefit while reducing harm from bias, misuse and unreliable conclusions.

CS-10-F-01
Responsible internet use and information security

Theory scope: Technology addiction; online information; privacy; spam, spyware and cookies; cloud security; protection methods and the purpose of technology laws.

Short questions | 3 marks each

S178. What is technology addiction? State two possible effects. [3 marks]

Technology addiction is a pattern of excessive or poorly controlled technology use that interferes with everyday life.

It can reduce concentration and study time.

It may also disrupt sleep, physical activity and relationships.

S179. Give two habits that support balanced use of digital devices. [3 marks]

Set reasonable study, recreation and screen-free times, including breaks.

Turn off unnecessary notifications and keep devices away during sleep or focused work.

Regular exercise and face-to-face activities help maintain balance.

S180. Define cybersecurity. Why is it needed? [3 marks]

Cybersecurity protects devices, networks and information from digital threats.

It helps prevent unauthorized access, theft, damage and disruption.

For example, secure accounts and updated software reduce the risk of data being stolen.

S181. Differentiate between privacy and data security. [3 marks]

Privacy concerns how personal information is collected, used and shared.

Data security concerns protecting information from unauthorized access, alteration or loss.

For example, limiting who may collect your details supports privacy, while encryption helps secure those details.

S182. What is spam? How can it affect a user? [3 marks]

Spam is unsolicited bulk messaging, often sent through email.

It wastes time and may contain scams, harmful links or unwanted advertising.

Spam filters and avoiding suspicious messages reduce exposure.

S183. What is spyware? State one risk it creates. [3 marks]

Spyware is software that secretly gathers information about a user or device.

It may record browsing activity, personal information or credentials.

Trusted software sources, security updates and anti-malware tools help reduce the risk.

S184. What are browser cookies? Give one useful purpose and one privacy concern. [3 marks]

Cookies are small pieces of data a website stores through the browser.

A useful purpose is remembering a session or user preference.

A privacy concern is that some cookies can support tracking across sites; users can review browser settings and permissions.

S185. What is phishing? State one way to avoid it. [3 marks]

Phishing uses deceptive messages or websites to trick people into revealing information or performing an unsafe action.

A fake bank message may ask for a password.

Avoid following suspicious links; open the service through a known address and verify unexpected requests independently.

S186. Give two ways the internet can be used to share helpful information. [3 marks]

The internet can distribute educational lessons and reliable study resources.

It can also share verified community notices or public-service information quickly.

Before posting, users should check accuracy and make the message clear for the intended audience.

S187. How can false information harm people? State one way to check it. [3 marks]

False information may cause fear, damage reputations or lead to poor decisions.

Check the claim against reliable, independent sources and verify its date and context.

Do not forward it simply because it is popular or emotionally convincing.

S188. Give two ways to protect information stored in the cloud. [3 marks]

Use a strong, unique password with multi-factor authentication where available.

Give files only the access permissions needed and avoid making private data publicly accessible.

Maintain appropriate backups and review sharing settings regularly.

S189. Differentiate between authentication and authorization. [3 marks]

Authentication checks who a user is, for example through a password and a second factor.

Authorization determines what the authenticated user is allowed to access or do.

A student may log in successfully but still be unable to edit a teacher's records.

S190. How do encryption and software updates protect data or devices? [3 marks]

Encryption changes readable data into protected form that requires a suitable key to read.

Updates fix known defects and security weaknesses in software.

Together they reduce exposure to data theft and attacks, although they do not replace careful access control.

S191. What is the purpose of digital privacy and cybercrime laws? [3 marks]

Digital privacy laws establish rules for collecting, using and sharing personal information.

Cybercrime laws define and address unlawful digital activities and provide routes for investigation or redress.

Users and organizations should respect these rules and use technology responsibly.

S192. How can social-media peer pressure lead to unsafe sharing? [3 marks]

A person may share private photos, location or passwords to gain approval or avoid exclusion.

Once shared, the information may be copied or misused.

Users should set boundaries, refuse unsafe requests and seek support when pressured.

CS-10-F-01 | Long answers

Long questions | 6 marks each

L074. Explain five strategies for safe and responsible internet use that protect personal information. [6 marks]

1. Secure accounts: use strong, unique passwords and enable multi-factor authentication where possible.

2. Recognize deception: avoid suspicious links, attachments and requests for passwords or verification codes.

3. Limit personal sharing: do not publicly reveal unnecessary identity, location or financial information.

4. Control access: review privacy settings and give only necessary permissions to people and applications.

5. Maintain devices: install trusted updates, use appropriate security software and back up important files.

6. Share responsibly: verify information, respect others' privacy and do not forward abusive content.

These actions reduce common risks while supporting useful internet activities such as learning, communication and access to services.

L075. Compare spam, spyware, phishing and tracking through cookies. Explain one protective action for each. [6 marks]

Issue	Meaning or risk	Protective action
Spam	Unsolicited bulk messages that may include scams.	Use filters and avoid interacting with suspicious messages.
Spyware	Software secretly collecting information.	Install trusted software, update systems and use anti-malware protection.
Phishing	Deception that tricks users into revealing information or taking unsafe actions.	Verify requests through a known, independent route.
Cookie-based tracking	Some cookies support monitoring browsing across sites.	Review browser controls and limit unnecessary tracking permissions.

Not every cookie is harmful: some maintain sessions or remember preferences. Security depends on understanding each threat and applying an appropriate defence, rather than treating all online activity as equally risky.

L076. Explain the effects of technology addiction on study, health and relationships. Suggest a balanced-use plan for a student. [6 marks]

Effects on study: excessive use reduces focused study time and makes attention harder to maintain.

Effects on health: late-night use may disrupt sleep, while long inactive periods reduce physical activity.

Effects on relationships: frequent distraction can reduce time and attention given to family and friends.

Balanced-use plan: set fixed study and leisure periods, take regular breaks and disable non-essential notifications.

Keep a screen-free period before sleep and keep the phone away during focused work.

Include exercise and face-to-face activities. Review the plan each week and seek support from a trusted person if technology use repeatedly interferes with daily life.

L077. Compare passwords, multi-factor authentication, encryption, firewalls and software updates as methods of protecting devices and data. [6 marks]

Passwords: protect accounts through a secret credential; unique, strong passwords reduce guessing and reuse risks.

Multi-factor authentication: requires another independent check, reducing the risk from a stolen password alone.

Encryption: protects the readability of data during storage or transmission, provided keys are managed securely.

Firewalls: apply rules to network traffic and help block unauthorized connections.

Updates: correct known security weaknesses in operating systems and applications.

Together these provide different layers of protection. None guarantees complete security by itself: users must also control permissions, avoid deception, maintain backups and monitor unusual activity.

L078. Explain key security and privacy concerns of cloud computing. Suggest measures involving account security, access permissions, backups and careful sharing. [6 marks]

Concerns: cloud data is remotely stored and accessed through accounts and networks. Weak credentials, excessive sharing, provider or service problems and unauthorized access may expose or disrupt information.

Account security: use strong unique passwords and multi-factor authentication.

Permissions: give users only needed access and review shared links regularly.

Backups: keep suitable recoverable copies of important information and understand available recovery features.

Careful sharing: check the intended audience before uploading or sharing personal data; use encrypted connections.

Also review the service's settings and responsibilities. A reputable service can still be misconfigured by its users, so secure use remains necessary.

L079. Explain how technology laws can support digital privacy and information security. Discuss responsible information sharing and respect for others' personal data. [6 marks]

Technology laws can define unlawful access, data misuse and other offences and provide mechanisms for complaints and investigation.

Privacy rules can set expectations for collecting, using and disclosing personal information and can make organizations accountable for improper handling.

Responsible sharing supports these goals: obtain appropriate permission before sharing personal details or images, verify facts and give credit for borrowed content.

Do not disclose passwords, private documents or sensitive information to an unnecessary audience.

Use secure storage and appropriate access controls. Legal protections and everyday security habits work together; the existence of a law does not itself stop every threat.

CS-10-F-02
Cyberbullying, cybercrime and protection

Theory scope: Harassment impacts and response; backups; access control; monitoring; public and private clouds; hardware/software security; reporting concepts.

Short questions | 3 marks each

S193. Define cyberbullying. Give one example. [3 marks]

Cyberbullying is using digital communication to repeatedly or deliberately harass, threaten, humiliate or harm someone.

Example: posting abusive comments about a classmate in a group.

It can occur through messages, games, social platforms or other online services.

S194. State two effects of cyberbullying on a student. [3 marks]

It can cause anxiety, fear, sadness or reduced confidence.

It may affect concentration, attendance, studies and relationships.

Victims should be supported and should not be blamed for the abuse.

S195. What should a victim do before deleting abusive messages? [3 marks]

Preserve evidence such as screenshots, message details, account names, links and dates.

Keep the evidence safely without forwarding it unnecessarily.

Then use blocking and reporting tools and seek help from a trusted adult or appropriate authority.

S196. Give two ways to support a friend who is being harassed online. [3 marks]

Listen calmly, take the problem seriously and reassure the friend that the abuse is not their fault.

Help preserve evidence and report the content or account.

Encourage support from a trusted adult, teacher or appropriate reporting channel.

S197. State two actions schools or parents can take to prevent cyberbullying. [3 marks]

Schools can teach respectful online behaviour and provide a clear, safe way to report incidents.

Parents can discuss online experiences and offer supportive guidance.

Both should respond promptly, protect affected students and avoid blaming the victim.

S198. Why should users block and report abusive accounts? [3 marks]

Blocking limits further contact from the abusive account.

Reporting alerts the platform so it can review the behaviour under its rules.

Users should preserve relevant evidence first and seek additional help if threats or serious harassment continue.

S199. What is a backup? Why is it needed? [3 marks]

A backup is a separate copy of important data kept for recovery.

It helps restore files after accidental deletion, device failure or an attack.

Backups should be protected and checked to ensure that restoration works.

S200. How do access permissions protect files? [3 marks]

Access permissions define which users can view, change or delete files.

Giving only necessary access reduces unauthorized disclosure and accidental or deliberate damage.

For example, students may be allowed to read results but not edit the marks database.

S201. Give two ways network monitoring can help detect threats. [3 marks]

Monitoring can detect unusual traffic, repeated failed logins or unknown connections.

Alerts help administrators investigate quickly and block suspicious activity.

It supports detection and response, but does not guarantee that every attack will be stopped.

S202. Differentiate between public and private clouds. [3 marks]

A public cloud is a provider-operated service offered to multiple customers with separated accounts or resources.

A private cloud is reserved for one organization and may be hosted internally or externally.

Both require authentication, appropriate permissions and other security controls.

S203. Give one hardware method and one software method used to protect a device. [3 marks]

Hardware method: use a physical lock or secure cabinet to prevent theft or tampering.

Software method: use a firewall or antivirus tool to help block or detect threats.

Physical and software controls address different risks and work best together.

S204. What is a cybercrime? Give two examples. [3 marks]

A cybercrime is an unlawful act involving computers, networks or digital information.

Examples include unauthorized account access and online fraud.

Serious digital harassment may also be addressed under applicable law and should be reported through appropriate channels.

S205. State the general steps for reporting an online offence while keeping evidence safe. [3 marks]

Preserve messages, screenshots, links, usernames and dates securely.

Report through the platform and seek help from a trusted adult or institution where relevant.

Use the current official reporting channel of the relevant authority for serious offences, and avoid editing or publicly spreading evidence.

S206. Why should users avoid replying aggressively or forwarding abusive content? [3 marks]

Aggressive replies may escalate the situation and expose the victim to more abuse.

Forwarding harmful content can increase its reach and cause further humiliation.

A safer response is to preserve evidence, block, report and seek support.

CS-10-F-02 | Long answers

Long questions | 6 marks each

L080. A student receives repeated threatening messages. Explain a safe response involving evidence, blocking, platform reporting, support from a trusted adult and formal reporting where necessary. [6 marks]

1. Preserve evidence: save threatening messages, usernames, links, dates and screenshots in a safe place.

2. Avoid escalation: do not threaten back or repeatedly argue with the sender.

3. Block and secure: block the account, review privacy settings and protect account credentials.

4. Report to the platform: use its reporting tools so the content and account can be reviewed.

5. Seek support: tell a trusted adult, parent, teacher or school safeguarding contact and ask for help.

6. Escalate serious threats: use the relevant authority's current official reporting channel; seek immediate local assistance if there is immediate danger.

The student should not be blamed for the harassment.

L081. Evaluate the roles of schools and parents in preventing cyberbullying. Suggest five practical measures. [6 marks]

Schools should teach digital citizenship, explain the effects of harassment and establish clear rules for school-managed platforms.

Parents should discuss respectful online behaviour and create an environment where children can ask for help without fear of blame.

Five practical measures:

1. Provide regular awareness sessions.

2. Set clear reporting and response procedures.

3. Encourage early, confidential reporting to a trusted adult.

4. Preserve evidence and respond promptly to incidents.

5. Offer emotional and academic support to affected students.

Cooperation matters: schools can address peer behaviour while parents provide support at home. Supportive guidance is more effective than ignoring the problem or punishing victims for reporting it.

L082. Explain how backups, access control, network monitoring, antivirus software and a firewall protect information. [6 marks]

Backups: separate, protected copies allow recovery after deletion, damage or ransomware. Restoration should be checked.

Access control: accounts and permissions limit who can view or change information.

Network monitoring: unusual connections, traffic or failed logins can trigger investigation and response.

Antivirus or anti-malware: scans for malicious software and helps detect or remove threats.

Firewall: filters network traffic according to rules and helps block unauthorized connections.

These controls address recovery, access, detection and prevention. They should be combined with updates, strong account security and careful user behaviour, because no single measure prevents every possible incident.

L083. Compare public and private clouds in ownership or access, control and suitability. Explain why security measures are necessary in both. [6 marks]

Aspect	Public cloud	Private cloud
Access/tenancy	Provider offers services to multiple customers with separated resources or accounts.	Dedicated to one organization.
Control	Customer controls allowed settings; provider manages shared infrastructure.	Organization has greater dedicated control, depending on hosting arrangements.
Suitability	Flexible access and capacity for many common services.	Organizations needing dedicated arrangements and tailored control.

A private cloud is not automatically safer, and a public cloud is not automatically insecure.

Both need authentication, permissions, encryption where appropriate, updates, monitoring and recovery arrangements. Security depends on design, configuration and operation as well as the hosting model.

L084. Describe common cybercrimes and the purpose of laws addressing them. Explain general reporting steps and the types of evidence a victim should preserve. [6 marks]

Common cybercrimes include unauthorized access to accounts or systems, online financial fraud and malicious interference with data or services. Applicable laws define offences and support reporting, investigation and accountability.

Reporting steps: stop unnecessary contact, preserve evidence, report the account or content on the platform, and use the relevant authority's current official reporting channel for serious offences.

Evidence may include original messages, screenshots, links, usernames, timestamps, transaction references and relevant records.

Do not alter evidence or publish private information unnecessarily. Seek help from a trusted adult or institution where needed, and record any complaint reference provided. Exact legal procedures depend on the applicable jurisdiction.

L085. Prepare a device-protection plan using both hardware and software methods. Explain five measures for a school computer laboratory. [6 marks]

1. Physical protection: secure the laboratory, lock unused equipment and control access to storage or network hardware.

2. Accounts and permissions: give students individual restricted accounts and reserve administrator rights for authorized staff.

3. Software protection: install trusted anti-malware tools and configure firewalls appropriately.

4. Maintenance and monitoring: apply updates, inspect unusual activity and remove unauthorized software through an approved process.

5. Recovery: maintain protected backups of important files and test restoration.

6. User guidance: teach students to avoid suspicious downloads, log out after use and report faults or threats.

Together, physical and software controls protect devices, information and service availability.

CS-10-F-03
The digital divide and accessible solutions

Theory scope: Unequal access; effects on critical information; inclusive application design; ways to reduce inequality; students' role.

Short questions | 3 marks each

S207. Define the digital divide. [3 marks]

The digital divide is the gap between people or groups in access to, skills for, and effective use of digital technology.

It can involve devices, connectivity, affordability and digital literacy.

The gap affects access to education, work and important information.

S208. State two causes of unequal access to digital technology. [3 marks]

One cause is cost: some families cannot afford suitable devices or internet service.

Another is limited infrastructure, such as weak network coverage or unreliable electricity.

Differences in skills, language and accessibility can also restrict effective use.

S209. How can poor internet access affect a student's education? [3 marks]

Poor internet access can prevent attendance at live lessons and delay downloading resources.

It can make submitting assignments and communicating with teachers difficult.

The student may therefore lose learning opportunities available to well-connected classmates.

S210. Give two ways the digital divide affects access to important information. [3 marks]

People without effective access may miss educational or job information.

They may also find it harder to obtain public-service updates or complete online applications.

These barriers can increase inequality in everyday opportunities.

S211. Why should an application work on low-cost devices? [3 marks]

Not all users can afford recent or powerful devices.

An application that works on low-cost phones reaches more people and reduces exclusion.

Efficient use of memory, storage and processing also improves performance on older equipment.

S212. State two features that help users with slow internet connections. [3 marks]

Provide lightweight text pages and compressed media so less data is needed.

Allow downloads or offline access for later use.

Also support resumable downloads or low-resolution media options where suitable.

S213. How can local-language content improve access to information? [3 marks]

Local-language content helps users understand information in a language they know.

It reduces the barrier created by unfamiliar language or technical wording.

For example, Urdu instructions can help users who find English-only services difficult.

S214. What is the role of accessibility features such as captions or screen-reader support? [3 marks]

Captions make audio information available to people who cannot hear it or cannot play sound.

Screen-reader support helps users with visual impairments access text and controls.

Accessible design therefore expands who can use an application effectively.

S215. Give two actions students can take to reduce the digital divide. [3 marks]

Students can share digital skills with classmates and community members.

They can help distribute permitted study resources in low-data or offline forms.

They may also support supervised device-donation or community-learning activities.

S216. Why should designers consider users' devices, connectivity and abilities? [3 marks]

Users differ in device capability, internet access, language and physical or learning needs.

Ignoring these differences may exclude people from essential information.

Designers should offer efficient, understandable and accessible ways to complete important tasks.

CS-10-F-03 | Long answers

Long questions | 6 marks each

L086. Explain causes and effects of the digital divide in education, employment and access to essential information. Suggest two solutions. [6 marks]

Causes: the cost of devices and data, limited coverage or electricity, low digital skills and inaccessible or unfamiliar-language content.

Education: students may miss lessons, resources and online assignments.

Employment: people may struggle to find vacancies, learn skills or submit applications.

Essential information: limited access can obstruct public notices, services and useful guidance.

Solution 1: schools and communities can improve shared access through suitable devices, connectivity and training.

Solution 2: services can provide low-data, offline, accessible and local-language options.

The digital divide is more than owning a device: people also need affordable access, suitable content and the skills to use it effectively.

L087. A learning application must serve students with old phones and limited mobile data. Suggest five design features that make it more accessible. [6 marks]

1. Lightweight pages: keep layouts and files small so old phones can load them.

2. Text-first lessons: offer readable text instead of requiring video for every topic.

3. Download/offline access: let students save lessons and practise without a continuous connection.

4. Media choices: provide compressed images, audio or lower-resolution video rather than forcing large downloads.

5. Inclusive interface: use clear controls, local-language options and accessible text and captions.

6. Reliable progress: save work locally when possible and synchronize when connectivity returns.

These features reduce device and data demands while helping more students reach the same learning material.

L088. Discuss how schools, communities and students can help reduce unequal access to technology and information. [6 marks]

Schools: provide shared computer access, digital-skills lessons and learning resources that also work with limited connectivity.

Communities: establish accessible learning spaces, support affordable connectivity and organize appropriate device-sharing or donation initiatives.

Students: help peers learn basic skills, share permitted resources in efficient formats and assist community users respectfully.

Designers and institutions should also provide local-language and accessible content so access is useful in practice.

The aim is to improve devices, connectivity, affordability, skills and inclusion together. A device alone does not solve the divide if the user cannot connect, understand the content or operate the service.

L089. Compare an application designed only for fast internet with one offering text, low-data access, local languages, captions and offline content. Explain how these choices affect inclusion. [6 marks]

Fast-internet-only design may require large videos, recent devices and a continuous connection. Users with expensive data, weak coverage or older phones can be excluded.

Inclusive design provides text alternatives, compressed media and low-data modes. Offline downloads allow study when a connection is unavailable.

Local-language content improves understanding. Captions make audio material accessible, while clear text and screen-reader support help users with different needs.

These choices let more people complete the same task rather than receive a reduced opportunity simply because of their circumstances.

Design should be tested with representative users, including those with slow connections and low-cost devices.

CS-10-G-01
Communicating ideas through digital media

Theory scope: Researching simple topics; extracting key ideas; copy and graphics; audience and format; YouTube, Instagram and Facebook; effective publication.

Short questions | 3 marks each

S217. What is digital communication? Give two examples of tools used for it. [3 marks]

Digital communication is the exchange of ideas and information using electronic devices and digital services.

Examples of tools include email and video-conferencing applications.

It can combine text, images, audio and video to communicate with different audiences.

S218. Why should a message be adapted to its audience? [3 marks]

Audiences differ in age, language, knowledge and purpose.

Matching wording, examples and format helps people understand and use the message.

For example, a student revision post should use simpler language than a specialist research discussion.

S219. What is meant by extracting the key ideas from information? [3 marks]

Extracting key ideas means identifying the most relevant points and removing unnecessary detail.

The main message should remain accurate and complete for its purpose.

For an event notice, key ideas include what the event is, when and where it happens, and how to join.

S220. State two features of effective copy for an event poster. [3 marks]

Effective copy clearly states the event's name, date, time, venue and any required action.

It uses brief, readable wording suitable for the intended audience.

A clear heading and contact or registration information make the poster useful.

S221. How can graphics make a message easier to understand? [3 marks]

Relevant graphics can illustrate a process, emphasize a main point or summarize information quickly.

They should be clear, labelled when needed and connected to the message.

Decorative or misleading graphics can distract, so images should be selected carefully.

S222. Choose a suitable platform for a long teaching video and give a reason. [3 marks]

YouTube is a suitable choice for a long teaching video.

It supports video-based explanation that viewers can pause and revisit.

A clear title, organized explanation and captions help the intended students use it effectively.

S223. Choose a suitable platform for a product-photo collection and give a reason. [3 marks]

Instagram is a suitable option for a product-photo collection.

Its image-focused posts can display several views or features with short captions.

The choice should still depend on where the target customers actually look for information.

S224. How can a community use a Facebook group or page to share information? [3 marks]

A community page or group can publish verified announcements, event details and useful resources.

Members can ask questions or receive updates from authorized organizers.

Clear moderation and privacy rules help keep communication accurate and respectful.

S225. State two good practices for reaching an audience on digital platforms. [3 marks]

Use a clear message and a suitable format for the intended audience.

Publish useful, consistent content with readable visuals and accurate descriptions.

Review audience feedback to improve future posts, without using misleading claims.

S226. Why should creators check accuracy and credit sources before publishing? [3 marks]

Checking accuracy prevents misinformation and protects trust.

Crediting sources acknowledges the original creator and helps readers verify information.

Creators should also respect copyright and permissions when using text, images or other media.

S227. How can captions and readable text improve digital communication? [3 marks]

Captions let people follow a video without hearing its audio.

Readable text with suitable size, contrast and spacing makes information easier to understand.

Both improve access for users with different abilities, devices and viewing conditions.

CS-10-G-01 | Long answers

Long questions | 6 marks each

L090. A school is announcing a science exhibition. Explain how to collect accurate details, choose key information, write clear copy and select suitable graphics for a digital poster. [6 marks]

1. Collect accurate details from the organizer: event name, purpose, date, time, venue and participation requirements.

2. Select key information: retain what viewers need to understand and attend the event.

3. Write clear copy: use a strong heading, short sentences and a clear action such as "Register by Friday."

4. Choose relevant graphics: use an image or illustration connected to science and avoid clutter.

5. Design for readability: use suitable contrast, text size, spacing and a clear order of information.

6. Check and publish: verify all details and permissions, then use the platform most likely to reach students and parents. Provide an appropriate contact for questions.

L091. Compare YouTube, Instagram and Facebook as tools for communicating ideas. Give a suitable audience and type of content for each. [6 marks]

Platform	Suitable content	Audience/use
YouTube	Longer teaching videos, demonstrations or discussions.	Viewers seeking detailed explanations.
Instagram	Images, carousels and short visual clips.	People browsing visual updates, products or quick ideas.
Facebook	Community announcements, discussion and shared resources.	Local or interest-based groups and communities.

The best choice depends on the actual audience rather than the platform name alone.

For example, a lesson can appear as a full YouTube explanation, a short Instagram summary and a Facebook group announcement linking to resources. Wording and media should match each format while keeping the underlying information consistent and accurate.

L092. Explain how the same study tip can be presented as a short video, an image post and a community announcement. Describe how wording and design should change for each format. [6 marks]

Shared idea: "Revise regularly instead of leaving all preparation until the last day."

Short video: explain one small revision routine verbally, show an example timetable and add captions. Keep the main point easy to follow.

Image post: use one clear heading and a few short steps, such as "Review, practise, check." Use readable text and a relevant graphic.

Community announcement: state the advice in plain text, explain its benefit briefly and link any useful resource.

Across all formats, adapt length and visual detail to the audience. Keep the advice accurate and avoid a crowded design or unsupported claims.

L093. Describe five good practices for publishing clear and responsible digital content, including accuracy, audience awareness, suitable media, readability and crediting sources. [6 marks]

1. Accuracy: verify names, dates, facts and claims before posting, and correct mistakes promptly.

2. Audience awareness: use language, examples and detail suitable for the intended viewers.

3. Suitable media: choose text, images, audio or video according to the message and platform.

4. Readability and accessibility: use clear layout, readable text, captions and useful descriptions where appropriate.

5. Source credit: acknowledge borrowed material and respect copyright, permission and privacy.

6. Responsible interaction: communicate respectfully and moderate harmful comments when necessary.

These practices make digital content easier to understand, more trustworthy and more useful to the people receiving it.

EN-10-H-01
Digital market and customer research

Theory scope: Quantitative and qualitative methods; digital research tools; question design; collecting and interpreting results; sampling bias; correlation and causation.

Short questions | 3 marks each

S228. What is market research? Why is it useful before starting a business? [3 marks]

Market research collects and analyses information about customers, competitors and demand.

It helps check whether a business idea solves a real need.

It can guide decisions about features, price and service arrangements before money is spent.

S229. Differentiate between quantitative and qualitative research. [3 marks]

Quantitative research collects numerical information, such as prices, counts and percentages.

Qualitative research explores opinions, experiences and reasons in words.

A survey rating can show how many customers are satisfied, while an interview can explain why they feel that way.

S230. Give one example of a quantitative question and one qualitative question for an online tutoring service. [3 marks]

Quantitative question: "How many rupees could you pay per month for online tuition?"

Qualitative question: "What is the main difficulty you face when studying online?"

The first produces numerical values; the second provides descriptive reasons or experiences.

S231. Name two digital tools for collecting customer opinions. [3 marks]

An online form or survey tool can collect structured customer responses.

A video-call or messaging tool can support interviews and open-ended feedback.

The choice should match the research question and the target customers' access needs.

S232. When is an interview more useful than a closed-question survey? [3 marks]

An interview is more useful when detailed experiences or reasons are needed.

The researcher can ask follow-up questions and clarify unclear responses.

A closed-question survey is better suited to counting or comparing predefined responses from many people.

S233. What is sample-selection bias? Give one example. [3 marks]

Sample-selection bias occurs when the people selected do not fairly represent the target population.

For example, surveying only students with fast internet may underestimate difficulties using an online learning service.

The resulting conclusions may not apply to all intended customers.

S234. Why may surveying only your friends give misleading results? [3 marks]

Friends may share similar ages, incomes, locations or opinions.

They may also give favourable answers to be polite.

A more representative sample should include a wider range of actual target customers and neutral questions.

S235. Rewrite this biased question neutrally: "How much do you love our excellent learning app?" [3 marks]

Neutral version: "How would you rate your experience with this learning app?"

Offer balanced choices such as very poor, poor, average, good and very good.

This removes the assumption that users love the app or consider it excellent.

S236. Differentiate between an open-ended and a closed-ended research question. [3 marks]

An open-ended question lets respondents answer in their own words, such as "What should we improve?"

A closed-ended question provides fixed choices, such as "Morning or evening?"

Open questions explore reasons; closed questions are easier to count and compare.

S237. What is the difference between correlation and causation? [3 marks]

Correlation means that two variables vary together.

Causation means that a change in one variable produces a change in another.

A correlation alone does not establish causation because other factors or selection effects may explain the relationship.

S238. A survey of 50 students finds that 30 prefer evening tuition. What percentage prefer evening tuition? [3 marks]

Percentage = (number preferring evening / total surveyed) x 100.

= (30/50) x 100 = 60%.

Therefore 60% of the surveyed students prefer evening tuition; this need not represent all students.

S239. How can social-media comments and reviews provide customer insights? [3 marks]

Comments and reviews reveal preferences, complaints and descriptions of real experiences.

A business can identify repeated themes, such as difficulty using a feature.

These findings should be checked carefully because commenters may not represent all customers.

S240. Why should research questions be clear and focused on one issue? [3 marks]

Clear questions are easier to understand and answer accurately.

Asking one issue at a time avoids confusion, such as combining price and teaching quality in one rating.

This produces results that are easier to interpret and use.

S241. How can survey results help a business choose a price or service time? [3 marks]

Responses can show how much customers can afford and which times they prefer.

The business can compare these findings with costs and practical constraints.

For example, strong evening demand may support an evening batch, but the sample should represent intended customers.

EN-10-H-01 | Long answers

Long questions | 6 marks each

L094. Plan digital market research for an affordable online tutoring service. State the research aim, target respondents, suitable tools, one quantitative question and one qualitative question. [6 marks]

Aim: find whether Class 9 and 10 students need affordable online tuition and what arrangements would suit them.

Respondents: include students from different schools, areas and levels of internet access, with appropriate consent arrangements.

Tools: use an online form for structured answers and short interviews for deeper explanations; provide another response method for people without suitable internet.

Quantitative question: "What monthly fee range could you afford?" Give balanced, non-overlapping ranges.

Qualitative question: "What is your biggest difficulty when learning online?"

Analyse numerical responses and recurring themes. Use the results to plan timing, price and support, while checking sampling bias and avoiding unnecessary personal-data collection.

L095. Compare quantitative and qualitative research. Explain when each is useful and how the two can be combined to understand customer needs. [6 marks]

Quantitative research measures numerical facts or responses, such as preferred times, affordable fees or satisfaction ratings. It helps count preferences and compare groups.

Qualitative research explores experiences and reasons through interviews or open-ended responses. It helps explain why customers have particular needs.

Combined example: a survey may show that 60% of respondents prefer evening tuition. Interviews may reveal that they are occupied with school or travel during the day.

The numerical result estimates the preference within the sample; the explanation guides a useful service design.

Both require suitable questions and respondents. More responses do not remove bias if the sample is unrepresentative.

L096. A business surveys only urban students who already own expensive phones. Explain the sampling bias and suggest ways to obtain a more representative sample for a nationwide study app. [6 marks]

The sample excludes rural students, students with low-cost phones and people with limited internet access. It may overestimate device capability, willingness to pay and ability to stream lessons.

Therefore results from these urban, well-equipped respondents should not be treated as national preferences.

Improve the sample by including students from different regions, income groups, school types and device/access levels.

Use more than one recruitment route and allow suitable offline or assisted responses so low-access students can participate.

Check how the final sample compares with the target population and report remaining limitations. Ask neutral questions and protect respondents' information.

L097. A survey of 100 students finds that 60 prefer evening lessons, 25 prefer afternoon and 15 prefer morning. Present the percentages, state two useful insights and identify one limitation of these findings. [6 marks]

Preferred time	Responses	Percentage
Evening	60	60%
Afternoon	25	25%
Morning	15	15%

Each percentage = responses/100 x 100. The percentages total 100%.

Insight 1: evening has the largest preference, so an evening class is worth considering.

Insight 2: 40% prefer another time, so one evening batch may not meet all needs; an additional option or recording could help.

Limitation: the result applies directly to the surveyed sample. If respondents were selected from only one school or access group, the findings may not represent all intended customers.

L098. Students who use an education app have higher marks in a survey. Explain why this does not prove the app caused the higher marks. Suggest two other possible factors and a more careful way to investigate. [6 marks]

The survey shows an association between app use and higher marks; it does not prove that the app caused the difference.

Possible factor 1: already motivated students may be more likely to use the app and study more.

Possible factor 2: users may also have better tuition, resources or prior achievement.

A more careful investigation would measure starting performance, compare similar groups and track change over time. Where appropriate and practical, a well-designed randomized study can reduce selection differences.

The analysis should account for other relevant factors and report limitations. A favourable correlation alone is not enough for a causal claim.

L099. Design four unbiased questions to investigate customer needs for a stationery delivery service. Include both quantitative and qualitative questions and explain how one response could guide a business decision. [6 marks]

1. Quantitative: "How many times did you buy stationery in the past month?"

2. Quantitative: "What delivery-fee range would you be willing to pay?" Provide balanced, non-overlapping choices including zero and an option to decline.

3. Qualitative: "What difficulties do you face when buying stationery?"

4. Qualitative: "What would make a stationery-delivery service useful to you?"

These questions do not assume the service is needed or already liked.

Decision example: repeated reports of travel difficulty could support home delivery, while affordable-fee responses help test whether delivery can cover its costs. Findings should be interpreted using a representative sample rather than only friends or existing supporters.